

ARIZONA GHOST TRAILS

by
Richard J. Hinton

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Title: Arizona ghost trails

Author: Richard J. Hinton

Release date: December 12, 2022 [eBook #69526]

Most recently updated: October 19, 2024

Language: English

Original publication: United States: Frontier Book Co., 1969

Other information and formats: www.gutenberg.org/ebooks/69526

Credits: Krista Zaleski and the Online Distributed Proofreading Team at <https://www.pgdp.net>

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GHOST TRAILS ***

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by
Richard J. Hinton

1969

Frontier Book Co., Publisher

Fort Davis, Texas 79734

From
The Handbook To Arizona

c1877

Limited to 1000 copies

SUMMARY OF MINING LAWS, FEDERAL, TERRITORIAL, AND LOCAL; DESERT AND TIMBER LANDS; HOMESTEAD AND PRE-EMPTION RIGHTS; SPANISH AND RAILROAD GRANTS. LAND OFFICES. OFFICERS OF THE TERRITORY. LEGISLATION ON IRRIGATION. ARTESIAN WELLS, MINING, ETC. ROUTES, DISTANCES, AND FARES FROM AND TO ALL THE PRINCIPAL POINTS EAST AND WEST, AND IN THE TERRITORY. ALTITUDE OF IMPORTANT POINTS. METEOROLOGICAL TABLES. MINERAL SPRINGS. SOUTHERN PACIFIC RAILROAD LANDS, ETC.

Before the passage of the Act of 1866, by Congress, the ownership of the mineral lands was retained by the nation. The first discussion of the

policy of selling such lands began in 1850, the argument being to make them a source of revenue. The policy of leaving the mineral land open for private exploration and development prevailed, and remained the rule until 1866. The uncertainty of titles, etc., was urged in 1865-6, as reason for a change. Under legislation preceding that date, no title could be or was conferred to mining claims, beyond possessory rights, maintained by working and payment of a small royalty. Citizens of the United States might explore and occupy under regulations as prescribed by law. In the absence of congressional enactment, local legislation was authorized to provide necessary rules; the local customs and district rules not in conflict with the United States laws were also recognized. The law was, in reality, a license only to go upon the mineral-bearing portions of the public domain. Ownership, however, attached to the minerals extracted, and the government had no claim to them, except so far as royalty or license fees were concerned.

The Act of 1872

Is not compulsory upon miners. They are not obliged to procure a United States patent for their claims. Those who do not, hold exactly the same relations that they did before its passage, provided no adverse claim is interposed. The Revised Statutes of the United States, Sections 2,318 to 2,352, of Title “Mineral Lands”; also, “Miscellaneous” provisions ditto, embracing Sections 910, 2,238, 2,258, 2,386 and 2,406, provide that for

Quartz Mines

Any person who is a citizen of the United States, or who has declared his intention to become a citizen, and no others, may locate and hold a mining claim 1,500 linear feet along the course of any mineral vein or lode subject to location; or any association of persons, severally qualified as above, may make joint location of such claim of 1,500 feet; but in no event can a location of a vein or lode, made subsequent to the date mentioned, exceed 1,500 feet along the course thereof, whatever may be the number of persons in the company.

With regard to the extent of surface ground adjoining a lode or vein, and claimed for the convenient working of the same, it is provided that the lateral extent of location, made after May 10th, 1872, shall, in no case, exceed 300 feet on each side of the middle of the vein at the surface, and that no such surface rights shall be limited by any mining regulations to less than 25 feet on each side of the middle of the vein at the surface, except where adverse rights, existing on the 10th of May, 1872, may render such limitations necessary; the end lines of such claims to be in all cases parallel with each other.

By the foregoing it will be seen that no lode-claim, located after May 10th, 1872, can exceed a parallelogram 1,500 feet in length by 600 in width, but whether surface ground of that width can be taken depends upon the local regulations, or State or Territorial laws then in force in the mining districts; but no such local regulations, or State or Territorial laws, shall limit a vein or lode-claim to less than 1,500 feet along its course, nor can surface rights be limited to less than 50 feet in width, unless adverse claims, existing on May 10th, 1872, render such lateral limitations necessary. It is provided by the Revised Statutes that the miners of each district may make rules and regulations not in conflict with the laws of the United States, or of the State or Territory in which the districts are situated, governing the location, manner of recording, and amount of work necessary to hold possession of a claim.

In order to hold a possessory right to a location made prior to May 10th, 1872, not less than \$100 worth of labor must be performed or improvements made thereon, within one year from the date of such location, and annually thereafter; in default of which the claim will be subject to re-location by any one else having the necessary qualifications, unless the original locator, his heirs, assigns, or legal representatives, have resumed work after such failure and before the re-location. The expenditures required upon mining claims may be made from the surface, or in running a tunnel for the development of such claims. The Act of February 11th, 1875, provided that where a person or company has run a tunnel for the purpose of developing a lode or lodes, the money so expended shall be considered as expended on the said lode, and the owner or owners shall not be required to perform work on the surface to hold the claim.

Individual proof of citizenship may be made by affidavit. If a company, unincorporated, by the agent's affidavit; if a corporation, by the filing of a copy of charter or certificate of incorporation. Locators against whom no adverse rights rested on the date of the Act of 1872, shall have, on compliance with general law and recognized custom, the exclusive right to

Possession and Enjoyment

Of the surface inclosure, and of "all veins, lodes, and ledges which lie under the top or apex of such lines, extended downward vertically," even though they in their descent extend outside the "side-lines of such surface locations." The right to such outside parts of veins or ledges is confined to all that lies between "vertical planes drawn downward," as described, so continued that these planes "will intersect" the exterior parts of the said "veins or ledges." The surface of another's claim cannot be entered by the locator or possessor of such lode or vein.

What Constitutes a Deposit.

The word “deposit” has always been construed by the Land Office to be a general term, embracing veins, lodes, ledges, placers, and all other forms in which the valuable metals have ever been discovered. Whatever is recognized as a mineral by standard authorities, where the same is found in quality and quantity sufficient to render the land sought to be patented more valuable on this account than for purposes of agriculture, is treated by the Land Office as coming within the meaning of the act. Lands, therefore, valuable on account of borax, carbonate of soda, nitrate of soda, sulphur, alum, and asphalt, it is held may be patented. The first section of the Act of 1872 says, “all valuable mineral deposits.” The sixth section uses the term “valuable deposits.” Deposits of fire-clay may be patented under the act, and so may iron deposits, which may be patented as vein or placer claims. Lands, more valuable on account of deposits of limestone, marble, kaoline, and mica than for purposes of agriculture, may be patented as mineral lands.

Miners' Form of Notice.

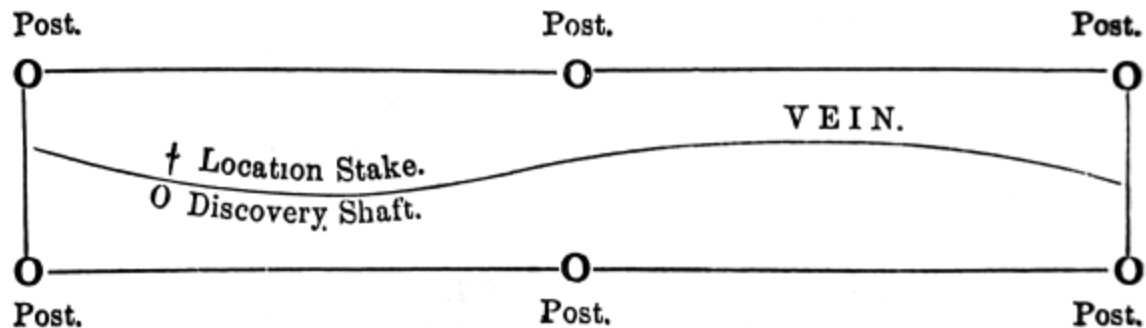
We hereby give notice that we have this —— day of —— A. D., 187-, located this, the ("Centennial") lode. We claim 1,500 feet in and along the vein, linear and horizontal measurement.

We claim 1,200 feet along the vein, running in a northwesterly course from the discovery shaft, and 300 feet running along the vein southeasterly from the discovery shaft. We also claim 150 feet on each side of the vein from center of crevice as surface ground.

W. —— M. ——, } Locators.
H. —— C. ——.

The Act of 1872 provides that no lode-claim can be recorded until after the discovery of a vein or lode within the limits of the ground claimed. The claimant should, therefore, prior to recording his claim, unless the vein can be traced on the surface, sink a shaft, or run a tunnel or drift to a sufficient depth therein to discover and develop a mineral-bearing vein, lode or crevice; should determine, if possible, the general course of such vein in the direction from the point of discovery, in which direction he will be governed in making the boundary of his claim on the surface; and should give the course and direction as nearly as practicable from the discovery shaft on the claim to some permanent, well-known points or objects, such as, for instance, stone monuments, blazed trees, the confluence of streams, etc., which may be in the immediate vicinity, and which will serve to perpetuate and fix the locus of the claim, and render it susceptible of identification from the description thereof given in the record of location in the district. He should drive a post, or erect a monument of stones at each corner of his surface ground, and at the point of discovery or discovery shaft, should fix a post, stake or board, upon which should be designated the name of the lode, the name or names of the locators, the number of feet claimed, and in what direction from the point of discovery; it being essential that the location notice filed for record, in addition to the foregoing description, should state whether the entire claim of fifteen hundred feet is taken on one side of the point of discovery, or whether it is

partly upon the other side thereof; and in the latter case, how many feet are claimed upon each side of such discovery point. The following diagram of surface boundaries, etc., of a lode, will aid the locator in this work:



Parties locating a lode are entitled to all the dips, spurs, angles, variations, and ledges of the lode coming within the surface ground.

The disordered condition of Arizona consequent on the Civil War and the continued hostilities of the Apaches, so impeded mining enterprises, compelling the abandonment of valuable mines and preventing full compliance with the conditions of the preceding acts, from no lack of diligence or skill on the part of miners, that some legislation seemed to be necessary to protect them from the injustice which a strict enforcement of the law would necessitate. The following acts were accordingly passed and approved on the dates specified.

An act approved March 1st, 1873, amends Section 5 of the Act of 1872, above referred to, so as to read as follows: "That the time for the annual expenditure on claims located prior to the passage of said act, shall be extended to the 10th day of June, 1874."

An act approved June 6th, 1874, made a further extension to January 1st, 1875.

An act approved February 11th 1875, so amends Section 2324, Revised Statutes, as to provide that where "a person or company has or may run a tunnel for the purpose of developing a lode or lodes owned by said person or company, the money so expended on said tunnel shall be taken and considered as expended on said lode or lodes, whether located prior to or since the passage of said act; and such person or company shall not be

required to perform work on the surface of said lode or lodes in order to hold the same, as required by said act.”

Recording Location.

Within a reasonable time, after the location shall have been marked on the ground, notice thereof, accurately describing the claim in manner aforesaid, should be filed for record with the proper recorder of the district, who will thereupon issue the usual certificate of location. The district regulations or customs are followed in this regard. Within ninety days after location, a location certificate must be filed in the office of the Recorder, in the county in which the lode is situated, which should be in the following form:

TERRITORY OF ARIZONA, } ss.
COUNTY OF ———

KNOW ALL MEN BY THESE PRESENTS, That ——— the undersigned, ha—— this ——— day of ——— A. D. 1877, located and claimed, and by these presents do locate and claim, by right of discovery and location, in compliance with the Mining Acts of Congress, approved May 19th, A. D. 1872, and all subsequent Acts, and with the local customs, laws and regulations, ——— feet, linear and horizontal measurement, on the ——— Lode, along the vein thereof, with all its dips, angles and variations, together with ——— feet on each side of the middle of said vein at the surface; and all veins, lodes, ledges and surface ground within the lines of said claim ——— feet, running ——— from center of discovery shaft. Said discovery shaft being situate upon said lode, and within the lines of said claim in ——— Mining District, county of ——— and Territory of Arizona, and further described as follows:

Said lode was located on the ——— day of ——— A. D. 1877.

ATTEST:

Date of certificate, _____ A. D. 1877.

Labor and Expenditures.

In order to hold the possessory rights to a claim of 1,500 feet of a vein or lode located as aforesaid, the Act requires that until a patent shall have been issued therefor, not less than \$100 worth of labor on the basis adopted by the local mining regulations shall be performed, or improvements made thereon, during each year; in default of which the claim will be subject to re-location by any other party having the necessary qualifications, unless the original locator, his heirs, assigns or legal representatives have resumed work thereon after such failure and before such re-location. The importance of attending to these details in the matter of location, labor and expenditure will be the more readily perceived, when it is understood that a failure to give the subject proper attention, may invalidate the claim.

Adverse Claims.

The seventh section of the United States laws provides for adverse claims; fixes the time within which they shall be filed to have legal effect, and prescribes the manner of their adjustment. An adverse mining claim must be filed with the register of the same land office with whom the application for patent was filed, or in his absence, with the receiver, and within the sixty days' period of newspaper publication of notice. It must be duly sworn to by the party or parties filing the adverse claim, and not by an attorney, before a officer authorized to administer oaths within the land-district, or before the register or receiver; fully set forth the nature and extent of the interference or conflict; whether the adverse party claims as a purchaser for valuable consideration, or as a locator; if the former, the original conveyance, or a duly certified copy thereof, should be furnished; or if the transaction was a mere verbal one he will narrate the circumstances attending the purchase, the date thereof, and the amount paid, which facts should be supported by the affidavit of one or more witnesses, if any were present at the time; and if he claims as a locator, he must file a duly certified copy of the location from the office of the proper recorder. It will be incumbent upon the adverse claimant to file a plat showing his claim, and its relative situation or position with the one against which he claims, so that the extent of the conflict may be the better understood. This plat must be made from an actual survey by a United States deputy surveyor, who will officially certify thereon to its correctness; and in addition there must be attached to such plat of survey a certificate or sworn statement by the surveyor, as to the approximate value of the labor performed or improvements made upon the claim of the adverse party, and the plat must indicate the position of any shafts, tunnels, or other improvements, if any such exist upon the claim of the party opposing the application. Upon the foregoing being filed within the sixty days as aforesaid, the register, or in his absence, the receiver, will give notice in writing to both parties to the contest that such adverse claim has been filed, informing them that the party who filed the adverse claim will be required within thirty days from the date of such filing to commence proceedings in a court of competent

jurisdiction, to determine the question of right of possession, and to prosecute the same with reasonable diligence to final judgment; and that, should such adverse claimant fail to do so, his adverse claim will be considered waived, and the application for patent be allowed to proceed upon its merits. When an adverse claim is filed as aforesaid, the register or receiver will indorse upon the same the precise date of filing, and preserve a record of the date of notifications issued thereon; and thereafter all proceedings on the application for patent will be suspended, with the exception of the completion of the publication and posting of notices and plat, and the filing of the necessary proof thereof, until the controversy shall have been adjudicated in court, or the adverse claim waived or withdrawn.

Tunnels.

Tunnels run for the development of a vein or lode, or for the discovery of mines, give the owner or owners the right of possession of all veins or lodes within 3,000 feet from the face of the tunnel to the same extent as if discovered from the surface, *i.e.*, 1,500 feet on the lode; and locations on the line of such tunnels of veins or lodes not appearing on the surface, made by other parties after the commencement of the tunnel, and while the same is being prosecuted with reasonable diligence, shall be invalid; but failure to prosecute the work on the tunnel for six months shall be considered an abandonment of the right to all undiscovered veins or lodes on the line of said tunnel. To avail themselves of the benefits of the law, the proprietors of a mining tunnel will be required to give a proper notice of their tunnel location, by erecting a substantial post, bound or monument at the face or commencement thereof, upon which should be posted a good and sufficient notice, giving the names of the party or parties claiming the tunnel right, the actual or proposed course or direction, height and width. At the time of posting notice and marking out the lines of the tunnel, as aforesaid, a full and correct copy of such notice and location must be filed for record with the mining recorder for the district.

Placer Claims.

The laws of the United States provide, also, that no location of a placer claim, made after July 9th, 1870, shall exceed 160 acres for any one person or association of persons, which location shall conform to the United States surveys. All placer claims located after May 10th, 1872, shall conform as nearly as practicable with the United States system of public surveys, and no such location shall include more than 20 acres for each individual claimant.

These provisions of the law are construed by the commissioner of the General Land Office, to mean that after the 9th of July, 1870, no location of placer claim can be made to exceed 160 acres, whatever may be the number of locators associated together, or whatever the local regulations of the district may allow; and that from and after May 10th, 1872, no location made by an individual can exceed 20 acres, and no location made by an association of individuals can exceed 160 acres; which location cannot be made by a less number than eight *bona fide* locators; but whether *as much* as 20 acres can be located by an individual, or 160 acres by an association, depends entirely upon the mining regulations in force in the respective districts at the date of the location; it being held that such mining regulations are in no way enlarged by the statutes, but remain intact and in full force with regard to the size of locations, in so far as they do not permit locations in excess of the limits fixed by Congress; but that when such regulations permit locations in excess of the maximum fixed by Congress, they are restricted accordingly. A local regulation is valid, therefore, which provides that a placer claim, for instance, shall not exceed 100 feet square. Congress requires no annual expenditures on placer claims, leaving them subject to the local laws, rules, regulations and customs.

District Mining Regulations.

The following will serve as a model for the framing of district laws. They will vary a little in detail, according to the requirements of the locality.

Bounds and Laws of —— District.

By virtue of a notice duly signed and posted, on —— 1877, a meeting of miners was held at place of posting notice on the —— inst., at which place and time the —— Mining District was formed. Mr. —— acted as Chairman, and —— acted as Secretary.

Following are the laws passed to govern the district:

1. The mining district shall be called the —— Mining District.
2. The district shall embrace the following described and bounded territory: commencing at the easterly end of the —— Mine, and running west of north along the eastern boundary of the —— Mining District to the western end of the —— Mine, in the —— Mining District, —— Mountains.
3. The County Recorder of —— County, by virtue of his office, shall be, ex-officio, recorder of this district.
4. Chapter six, of title thirty-two, revised statutes of the United States, is adopted as this article.
5. In the location of mines in this district, copies of the notices of location must be placed on the mines before any legal record of the same can be made by the recorder. Any location not so made shall be null and void.
6. All location notices must be filed in the office of the recorder within thirty days after the actual date of location.
7. The County Recorder shall be entitled to a fee of two dollars for each and every notice recorded by him.
8. The records of —— County are hereby adopted as the bona fide records of this district.
9. The annual meeting of the voters of this district shall take place and be holden on the first Monday in May in each year.

10. Ten days prior to the date of holding the annual meeting, the Recorder shall place, or cause to be placed or posted, in three of the most conspicuous places in said district, a notice stating the time when, and the place where, said meeting shall be holden, and shall designate in such notice that the meeting shall be holden for the purpose of transacting all and every kind of business which may be properly brought before it.

11. At each annual meeting, the voters of the district shall elect their chairman and secretary, who shall hold office for one year, or until their successors are appointed.

12. The chairman and secretary of this meeting shall hold office from the —— day of —— for one year, or until their successors are elected.

13. The secretary of each meeting shall keep full and complete records of the minutes and proceedings of their respective meetings, and cause the same to be placed on record in the office of the County Recorder.

14. These rules, regulations and by-laws shall not be altered, or in any way changed, except at a regular annual meeting of the miners of said district, and then only by a legal vote of two-thirds of all the voters present and voting.

15. Any and all persons who are citizens of the United States of America, or who have declared their intention to become such, and own shares of stock or interests in any mine in the district, or who has worked in any mine in the district for the twenty days preceding such meeting, shall be considered a legal voter, and entitled to vote at a miners' meeting.

16. Five dollars per day shall be allowed for each and every eight hours' work performed upon a mine for the purpose of holding title, or performing the necessary amount of work for a patent, and no other expenses shall be considered as expended for the purpose of holding or perfecting title.

17. All mines hereafter located in this district shall be marked by end and corner monuments or stakes, at least eighteen inches in height above ground, with sufficient marks placed in or upon them to show which end or corner of the claim they designate; if stakes are used, they must be sunk at least six inches in the ground, and have a blaze and figures upon one side.

18. All locations made and recorded previous to the adoption of these rules, regulations and by-laws are hereby legalized, so far as they may not

conflict with the same.

19. These rules, regulations and by-laws shall be filed and recorded in the office of the County Recorder of —— County, and shall be in full force and effect from and after this —— day of —— 1877.

20. The above proceedings, and the proceedings of any subsequent meeting, shall be signed by the chairman and secretary, and transmitted by them to the County Recorder without delay.

[Signed.]

—— ——— Miner.

—— ——— Miner.

—— ——— Miner.

—— ———

—— ———

I certify that the foregoing is a correct statement of the proceedings had, and of the laws adopted for the —— Mining District, this —— day of —— 1877.

—— ——— Secretary.

Water Rights.

The United States Revised Statutes provide:

1. That as a condition of sale in absence of legislation by Congress, the legislature of a State or Territory may provide rules for working mines, involving easements, drainage, and other necessary conditions; these to be expressed in the patent.

2. That all prior rights, arising from possession, in the use of water, and recognized by local laws, etc., or judicial decisions, shall be regarded as vested, and shall be protected. This right of way is also granted and confirmed. Damages are to accrue if a land-settler's rights are interfered with.

3. All land patents shall be subject to vested and accrued water rights, including ditches and reservoirs.

Officers of United States Land Offices are required to file with the General Land Office, the local laws on such matters. The following is a summary of those passed by the legislature of Arizona.

WATER RIGHTS IN THE TERRITORY.—All rivers, creeks, and streams of running water in the Territory of Arizona are deemed public, and applicable to the purposes of irrigation and mining. All the inhabitants of the Territory who own or possess arable or irrigable lands shall have the right to construct public or private acequias, and obtain the necessary water for the same from any convenient river, creek, or stream of running water.

All damages arising from construction of the acequias shall be assessed by the Probate Judge of the county in a summary manner.

No inhabitant of the Territory shall have right to erect any dam or build a mill, or place any machinery, or open any sluice, or make any dyke, except such as are used for mining purposes, or the reduction of metals, that may impede or obstruct irrigation.

When any ditch or acequia shall be taken out for agricultural purposes, the person or persons so taking out such ditch or acequia shall have the

exclusive right to the water, or so much as may be necessary for such purpose; and it at any time the water so required shall be taken for mining purposes, the damages shall be assessed and paid.

All owners and proprietors of arable or irrigable land bordering on, or irrigable by, any public acequia, shall labor on such public acequia, whether such owners or proprietors cultivate the land or not; and all persons interested in a public acequia, whether owners or lessees, shall labor thereon in proportion to the amount of the land owned or held by them, and which may be irrigated or subject to irrigation.

In all districts or precincts, the owners or proprietors of land irrigated by public acequias are annually called together by the Justices of the Peace, to elect one or more overseers for the acequias—and it is the duty of said overseers to superintend the opening, excavations and repairs of said acequias; to apportion the number of laborers furnished by the owners and proprietors; to regulate them according to the quantity of land to be irrigated by each one from said acequia; to distribute and apportion the water in proportion to the quantity to which each one is entitled according to the land cultivated by him; and in making such apportionment, he shall take into consideration the nature of the seed sown or planted, the crops and plants cultivated; and to conduct and carry on such distribution with justice and impartiality.

If any owner or proprietor of land irrigated by such acequia shall neglect or refuse to furnish the number of laborers required by the overseer, he shall be fined, and all fines shall be applied to the benefit of said acequia.

Water privileges are, since the United States Act of May 10th, 1872, located in the same manner as mines, subject to local regulations, i.e. by definitely locating the five acres by monuments, and recording with the District or County Recorder. If the local rules and decisions of the Courts make the privilege forfeitable for non-use, another party may come in and claim the water right.

The Federal Courts have decided that the right of way to construct flumes or ditches, over the public lands, is unquestioned. It has also been decided that the miners' right to water, within "reasonable limits," is not to be questioned. "It must be exercised," however, with due regard to the

general condition and needs of a community, and cannot vest as an individual monopoly.

Mill Sites.

Land non-mineral in character, and not contiguous to the vein or lode, used by the locator and proprietor for mining or milling purposes, can be included in any application for patent, to an extent not to exceed five acres, and subject to examination and payment as fixed for the superficies of the lode. The owner of a quartz mill or reduction, not a mine owner in connection therewith, may also receive a mill-site patent. Such sites are located under the mining act, and in compliance with local law and customs as recognized. Such possessory rights give title also to all growing timber thereon. There must in every case be given satisfactory proof of the non-mineral character of the site, and the improvements thereon must be equal to \$500 in value. A mill passes to a railroad, if located after a land grant inured to the road.

Homestead and Pre-emption.

HOMESTEADS.—Every head of a family, widow, single man or woman of the age of twenty-one years, who is a citizen of the United States, or who has declared his or her intention of becoming so, can enter upon 80 acres of government land within the limits of a railroad grant, or 160 acres outside said limits; and after a continuous residence upon it and cultivation for five years, an absolute title to the land will be given by the United States government, at a total cost of about \$9 on 80 acres, or \$18 on 160 acres.

SOLDIER'S HOMESTEAD.—Any soldier or sailor who served during the rebellion not less than 90 days, and was honorably discharged, can homestead 160 acres, either within or outside of the limits of a land grant, and his term of service will be deducted from the five years' residence required upon the land; but in any event he must reside one year upon it. Thus, if he served three years, he would have to reside upon the land two years; and in the event of his having served four or five years, one year's residence would be necessary.

A soldier or sailor has the privilege of filing application for homestead upon the land through an agent or attorney, and need not for six months commence actual settlement upon it. Absence from a homestead at any time, for more than six months, works a forfeiture of right to the land.

PRE-EMPTIONS.—Any person qualified under the homestead laws can pre-empt 160 acres of government land within the limits of a railroad grant, and after an actual residence upon and cultivation of the same for at least six months, can obtain title by payment of \$2.50 per acre, or, if outside the limits, \$1.25 per acre. It is imperative, however, that the person so pre-empting shall (with his family, if any) reside upon the land. The cultivation of a few acres is sufficient. The same person, after having complied with the requirements of the laws of pre-emption, can homestead 80 acres within the railroad grant, or 160 acres outside the limits. In this way, a soldier or sailor can secure 320 acres within the limits; and it is open to the *world at large* for *any* man to acquire his 240 acres.

AN ADDITIONAL HOMESTEAD.—In addition to the Homestead and Pre-emption laws, a recent act has been passed, whereby every settler, as the fruits of his industry, can obtain another freehold of 160 acres under the following act:

An Act to amend an Act entitled “An Act to encourage the growth of timber on western prairies.”

“Any person who is the head of a family, or who shall have arrived at the age of twenty-one years, and is a citizen of the United States, or who shall have filed his declaration of intention to become such, who shall plant, protect and keep in a healthy, growing condition for eight years, 40 acres of timber, the trees thereon not being more than twelve feet apart each way, on any quarter section of any of the public lands of the United States, or 20 acres on any legal subdivision of 80 acres, or 10 acres on any legal subdivision of 40 acres, or one-fourth part of any fractional subdivision of land less than 40 acres, shall be entitled to a patent for the whole of said quarter section, or of such legal subdivision of 80 or 40 acres or fractional subdivision of less than 40 acres, as the case may be, at the expiration of the said eight years, on making proof of such fact by not less than two credible witnesses.”

HOW TO PRE-EMPT.—When you have selected the land you wish for pre-emption or homesteading under whatever right, it is better to get a land attorney or clerk in the nearest land office to make out the necessary papers. This saves time, and the danger of mistakes.

THE LATEST REGULATIONS.—Commissioner Williamson, of the General Land Office, has issued a circular to all registers and receivers throughout the country, containing instructions requisite to carry into effect two Acts of Congress, approved on the 3rd of April, relative to homestead entries. The first provides a new method of making the final proof in homestead entries. It dispenses with the present necessity of attendance at the district land office. The person desiring to avail himself thereof must appear with his witnesses before the judge of a court of record of the county and State, or district and Territory in which the land is situated, and there make the final proof required by law according to the prescribed forms; which proof is required to be transmitted by the judge or the clerk of the court, together with the fee and charges allowed by law. The judge being absent in any

case, the proof may be made before the clerk of the proper court. The fact of the absence of the judge must be certified in the papers by the clerk acting in his place. If the land in any case is situated in an unorganized county, the statute provides that the person may proceed to make the proof in the manner indicated, in any adjacent county in the State or Territory. The fact that the county in which the land lies is unorganized, and that the county in which the proof is made is adjacent thereto, must be certified by the officer. The other law to which attention is invited by this circular is entitled "An Act for the relief of settlers on the public lands under the pre-emption laws." Under this statute, a person desiring to change his claim under a pre-emption filing to that of a homestead entry, should be required, on making the change, to appear at the proper land office with his witnesses, and show full compliance with the pre-emption law to the date of such change, as has heretofore been required in transmutation cases. Proof of such compliance must be forwarded with the entry papers to this office. When the person applies to make final proof, he must show continued residence and cultivation as required by the homestead law. In case an adverse claim has attached to the land, due notice in accordance with rules of practice must be given all persons in interest, of time and place of submitting proof in support of the application to make such change. The adverse claimants will be entitled to the privilege of cross-questioning the applicants' witnesses, and of offering counter proof.

Lands formerly designated

As Mineral,

Can be entered by preëmption upon proof that mines or minerals are not contained therein. Lands found, after entry as agricultural, to contain valuable mineral deposits, such entry will be cancelled. Where, however, a patent has issued, and the land has been afterwards found to embrace a valuable deposit or lode, the title is valid, as the land has ceased to be part of the public domain. Proof, however, that the deposit, lode or mine was known before the patent issued will invalidate title thereto. Titles to town sites and lots are held subject, also, to mineral rights, which remain in the United States.

Under Act of 1876, it was permitted to any person, under the limit of citizenship, or declaration of intent, to proceed upon the public land, and occupy such area, to the extent of one section, or 640 acres, which cannot be cultivated or used for agricultural purposes, with the artificial conveying of water thereon and irrigation; three years being given to construct the necessary works and improvements. The price of such land is to be \$1.25 per acre, one-fifth being required to be paid at the time of location. In consequence of doubt as to the character of land which this act was designed to embrace, and charges made of fraudulent entries, further legislation will doubtless be had on this subject.

Coal lands are allowed to be entered in legal subdivision parcels, not to exceed 160 acres to any one person, or double that quantity to an association; the price of the same to be \$20 and \$10 per acre, according to whether or not the same be located within fifteen miles of a completed railroad.

Rivers are deemed navigable only when they are used, or are susceptible of being used, for commercial highways. The shores and soil under them were reserved to the States respectively, and new States have the same jurisdiction and sovereignty as old ones. The Land Office has never permitted a complication of such rights by attempting or permitting the sales of any portion of the beds of said rivers or streams.

Saline lands are not subject to homestead or preëmption entry. This policy has been uniform since the beginning of our land system. The Supreme Court has held uniformly that Congress has uniformly designed to prevent the sale of saline deposits and springs. The existence of such deposit or spring withdraws any quarter or other large portion of a section from settlement and location.

United States Land Office Fees.

United States Land Office Registers and Receivers are permitted by law to charge the following fees:

Homestead or pre-emption declaratory statement	\$1.00
On final certificate for each 160 acres	5.00
” ” ” 320 ”	10.00
” ” ” section, or 640 acres	15.00
Locations by States under grants, for each 160 acres	1.00
For superintending public land sales	5.00
For acting on application for patent or adverse mineral claim	5.00
For testimony either in mineral or agricultural land cases, taken in writing, for claimants, each 100 words	.15

Under the laws of Arizona the County Recorders are authorized and required to keep a record of all mines and mineral deposits that are located. For this work they are entitled to receive for recording each claim:

Not to exceed one folio	\$1.00
For each additional folio	.20

It is also provided by act of territorial legislature, approved November 9th, 1864, that persons in the military service of the United States may locate mineral claims, all local or district regulations to the contrary notwithstanding.

Under the Act of December 30th, 1865, in relation to placer mines and mining, it is provided that in the county of Yuma, persons who in locating placers shall place, for the purpose of mining thereon, a pump or pumps with a capacity of 100 gallons per minute, may be entitled to locate of

placer land not to exceed 160 acres. This privilege is not to include placer land which can be worked by water brought in ditches or flumes.

Under Act of September 30th, 1867, it is provided that joint mining claims may be segregated, when any of the owners thereof refuse or fail to join in working them, after notices in the county or other newspaper published nearest thereto, for the period of four weeks. After such notice, the parties issuing may apply to the District Court; notice is then posted conspicuously by the clerk, for requiring the delinquents to appear within sixty days, and show why the prayer should not be granted. At the end of this last period two commissioners may be appointed, who choose a third; and they examine and report in writing. A decree shall issue in accord with the report. Thirty days are allowed for an appeal to the Supreme Court.

All grants of lands within the Territory, individual or corporate, whether held under Mexican or United States titles, must be recorded in the office of the County Recorder where situated. If not so entered, they are declared null and void. It is provided also that settlers shall be protected in the occupancy, use and improvement of 340 acres of public lands.

Arizona Mine Mills.

Within the past few months there have been brought into Arizona the following quartz mills, all of which are now being set up, or are already in operation:

Champion (steam drop)	2 stamps
Silver King	5 "
Townsend & Co. (5 already up)	10 "
Peck (originally Black Warrior)	10 "
Signal	10 "
McCrackin	20 "
Hackberry	10 "
Walnut Grove (10?)	5 "
Dean	10 "
Knowles (Empire Flat)	10 "
Masterson's, Turkey Creek	2 "
No. of stamps not mentioned.	
	——— ———
Total	94 stamps

Of the mills which have been in operation since and before last spring, we can recall the following:

Ostrich (it may be 10)	5 stamps
Morrill & Ketchum	6 "
Tidwell	3 "
Aztlan	5 "

Frederick's	10	”
Constancia (now Luke's)	10	”
Mineral Park	5	”
Crook	5	”
Greenwood	10	”
Bill Smith's	10	”
Total	60 stamps	

Yuma Sentinel, October.

APPENDIX

Table Showing the Value of any Amount of Gold Dust, from 1 grain to 10 ounces, at \$16 to \$23 per ounce.

OUNCES.								
No.	\$16.00 per oz.	\$17.00 per oz.	\$18.00 per oz.	\$19.00 per oz.	\$20.00 per oz.	\$21.00 per oz.	\$22.00 per oz.	\$23.00 per oz.
1	16	17	18	19	20	21	22	23
2	32	34	36	38	40	42	44	46
3	48	51	54	57	60	63	66	69
4	64	68	72	76	80	84	88	92
5	80	85	90	95	100	105	110	115
6	96	102	108	114	120	126	132	138
7	112	119	126	133	140	147	154	161
8	128	136	144	152	160	168	176	184
9	144	153	162	171	180	189	198	207
10	160	170	180	190	200	210	220	230
PENNYWEIGHTS.								
1	80	85	90	95	100	105	110	115
2	160	175	180	190	200	210	220	230
3	240	255	270	285	300	315	330	345
4	320	340	360	380	400	420	440	460
5	400	425	450	475	500	525	550	575
6	480	510	540	570	600	630	660	690
7	560	595	630	665	700	735	770	805
8	640	680	720	760	800	840	880	920

9	720	765	810	855	900	945	990	1035
10	800	850	900	950	1000	1050	1100	1150
GRAINS.								
1	3 $\frac{1}{3}$	3 $\frac{1}{2}$	3 $\frac{3}{4}$	4	4 $\frac{1}{6}$	4 $\frac{1}{3}$	4 $\frac{1}{2}$	4 $\frac{3}{4}$
2	6 $\frac{2}{3}$	7	7 $\frac{1}{2}$	8	8 $\frac{1}{3}$	8 $\frac{2}{3}$	9	9 $\frac{1}{2}$
3	10	10 $\frac{1}{2}$	11 $\frac{1}{4}$	12	12 $\frac{1}{2}$	13	13 $\frac{1}{2}$	14 $\frac{1}{4}$
4	13 $\frac{1}{3}$	14	15	16	16 $\frac{2}{3}$	17 $\frac{1}{3}$	18	19
5	16 $\frac{2}{3}$	17 $\frac{1}{2}$	18 $\frac{3}{4}$	20	20 $\frac{5}{6}$	21 $\frac{2}{3}$	22 $\frac{1}{2}$	23 $\frac{3}{4}$
6	20	21	22 $\frac{1}{2}$	24	25	26	27	28 $\frac{1}{2}$
7	23 $\frac{1}{3}$	24 $\frac{1}{2}$	26 $\frac{1}{4}$	28	29 $\frac{1}{6}$	30 $\frac{1}{3}$	31 $\frac{1}{2}$	33 $\frac{1}{4}$
8	26 $\frac{2}{3}$	28	30	32	33 $\frac{1}{3}$	34 $\frac{2}{3}$	36	38
9	30	31 $\frac{1}{2}$	33 $\frac{3}{4}$	36	37 $\frac{1}{2}$	39	40 $\frac{1}{2}$	42 $\frac{3}{4}$
10	33 $\frac{1}{3}$	35	37 $\frac{1}{2}$	40	41 $\frac{2}{3}$	43 $\frac{1}{3}$	45	47 $\frac{1}{2}$

Out of a ton of ore from the Stonewall Jackson Mine, adjoining the General Lee, in the Globe District, (Arizona) there was extracted October 25th, by the Pacific Refinery, San Francisco, ten bars of silver valued at \$3,800, which is over 36 per cent. metallic copper.

At Clinton, Arizona, the great copper mining center, the Longfellow Company have two furnaces running, and turn out as high as 10,000 pounds of pig copper daily. The furnaces used are Bennett's patent. Arizona has, without doubt, the richest and most extensive copper mines in the world. This metal is all shipped east by way of El Moro.

The Ores of Gold and Silver.

NAME.	COMPOSITION.
Native gold	Gold; silver up to 40 per cent.; copper and iron.
Quicksilver	Mercury, with sometimes a little silver.
Amalgam	Silver, 26 to 35; mercury, 74 to 65.
Arquerite	Silver, 87; mercury, 13.

Gold Amalgam	Mercury, 58 to 61; silver, 0 to 5; gold, 38 to 42.
Native silver	Silver, alloyed with other metals.
Bismuth	Bismuth, 27; lead, 33; silver, 15; iron, 4; copper, 1; sulphur, 16.
Native copper	Pure copper, with small quantity of silver through it.
Tellurium	Tellurium, with gold and iron, varying quantities.
Antimony	Antimony, containing at times silver, iron, or arsenic.
Discrasite	Antimony, 23; silver, 77.
Silver glance	Sulphur, 13; silver, 87.
Naumannite	Selenium, 27; silver, 73.
Eucairite	Selenium, 32; copper, 25; silver, 43.
Hessite	Tellurium, 37; silver, 63.
Stromeyrite	Sulphur, 16; silver, 53; copper, 31.
Sylvanite	Tellurium, 56; gold, 28; silver, 16.
Nagyagite	Tellurium, 13 to 32; lead, 51 to 61; gold, 6 to 9.
Sternbergite	Sulphur, 34; silver, 32; iron, 34.
Miargyrite	Sulphur, 21; antimony, 43; silver, 36.
Pyrargyrite, or ruby silver	Sulphur, 18; antimony, 23; silver, 59.
Proustite, or ruby silver	Sulphur, 20; arsenic, 15; silver, 65.
Freieslebenite	Sulphur, 19; antimony, 27; lead, 30; silver, 24
Tetrahedrite, or gray copper	Sulphur, arsenic, antimony, silver, copper, iron, zinc, and mercury, in most varying proportions.
Polybasite	Sulphur, 16; antimony, 13; silver, 71.
Stephanite	Sulphur, 16; antimony, 14; silver, 70.
Xanthocone	Sulphur, 21; arsenic, 15; silver, 64.
Fireblende	Sulphur, antimony, with silver, up to 62 per cent.

Cerargyrite, or horn silver	Chlorine, 25; silver, 75.
Embolite	Chlorine, 13; bromine, 20; silver, 67.
Megabromite	Chlorine, 9; bromine, 27; silver, 64.
Mikrobromite	Chlorine, 18; bromine, 12; silver, 70.
Bromyrite, or bromic silver	Bromine, 43; silver, 57.
Iodyrite, or iodic silver	Iodine, 54; silver, 46.
Jalpaite	Copper and silver glance.
Acanthite	Sulphuret of silver.
Crookesite	Copper, thallium, silver, and selenium.

Charles P. Stanton, geologist, writes to the Prescott “Miner,” under date of Nov. 9th, 1877, as follows: The great carboniferous basin of Arizona—and in all probability of the world—exists within 140 miles of Prescott. This immense coal deposit makes its first appearance in Southwestern Colorado and Northwestern New Mexico; but its great nucleus extends from Tierra Ausarilla, in Rio Arriba, New Mexico, to the Colorado River, a distance of 276 miles, and from the San Juan River, a distance of 138 miles. This immense area of 38,088 square miles is one continuous mass of coal. It lies between the parallels of 100 and 112 west longitude, and 35 and 37 north latitude, and all in Yavapai County.

Weight and Specific Gravity of the Common Minerals.

NAME.	COMPOSITION.	Specific gravity.	Weight of cubic inch (ounces).	Weight of cubic foot (pounds).
Gold, pure		19.3	11.169	1,206.25
Gold, native	Gold, silver, copper, iron	17.5	10.128	1,093.75
Silver, pure		10.6	6.134	662.50

Silver, native	Silver and other metals	10.	5.787	625.00
Copper, native	Copper	8.9	5.150	556.25
Vitreous Copper	Sulphuret of copper, 21; copper, 79	5.6	3.298	350.00
Copper pyrites	Sulphur, 35; copper, 35; iron, 30	4.2	2.430	262.50
Red copper	Copper, 89; oxygen, 11	5.9	3.414	368.75
Gray copper	Sulphur, arsenic, antimony, silver, copper, iron, zinc, and mercury in most varying proportions	4.8	2.777	300.00
Malachite	Carbonic acid, 20; oxide of copper, 72; water, 8	3.8	2.199	237.50
Tin oxide	Oxygen, 22; tin, 78	6.7	3.877	418.75
Tin pyrites	Tin, 27; copper, 30; iron, 13; sulphur, 29	4.4	2.546	275.00
Lead, pure		11.4	6.597	712.50
Galena	Sulphur, 13; lead, 87	7.5	4.340	468.75
Carb. of lead	Carbonic acid, 16; oxide of lead, 84	6.4	3.715	400.00
Zinc		7.0	4.051	437.50
Red oxide of zinc	Zinc, 80; oxygen, 19; oxide of manganese	5.4	3.125	337.50
Antimony		6.8	3.935	425.00
Gray oxide of antimony	Sulphur, 29; antimony, 71	4.5	2.025	281.25
Nickel		8.4	4.861	525.00

Arsenical nickel	Arsenic, 56; nickel, 44	7.5	4.340	468.75
Sulphuret	Nickel, sulphur, arsenic	6.2	3.588	387.50
Cobalt pyrites	Cobalt, 58; sulphur	5.0	2.893	312.50
Iron, cast		7.2	4.166	450.00
Iron, wrought		7.78	4.496	486.25
Iron pyrites	Sulphur, 53; iron, 47	4.9	2.835	306.25
Magnetic iron	Oxygen, 28; iron, 72	5.1	2.951	318.75
Arsenical iron	Arsenic, 46; sulphur, 20; iron, 34	5.7	3.298	356.25
Specular iron	Oxygen, 30; iron, 70	4.9	2.835	306.25
Hematite	Oxide of iron, 86; water, 14	4.0	2.314	250.00
Uranium, or pitch-blende	Oxygen, 15; uranium, 85	7.0	4.051	437.50
Baryta or heavy spar		4.0	2.314	250.00
Lime rock		3.0	1.736	187.50
Calc spar		2.7	1.562	168.75
Fluor spar		3.15	1.822	196.87
Quartz		2.69	1.673	167.452
Granite		2.78	1.608	173.75

NOTE.—A vein of ore one inch thick, six feet long, and six feet high, will measure three cubic feet; two inches, six cubic feet, and so on in proportion, allowing three cubic feet for every inch of ore in the lode, six feet high and six feet long.

Excellent Advice to the Emigrant Traveler

Is given by the California Immigrant Union, No. 248 Montgomery Street, San Francisco, which, if heeded, will be of service:

1. Buy your tickets for passage on railroad or steamboat, only at the office, before starting. Many of the runners who offer tickets for sale in the streets are swindlers. If you intend to go in a steamer or ship, examine the vessel before getting your ticket, and engage a particular berth or room in a part of the vessel that is clean, well ventilated and just comfortably warm.

2. Never show your money nor let any stranger know that you have any. Thieves prefer to rob emigrants, who generally carry money with them, and cannot stop to prosecute them, and have no acquaintances to aid in the prosecution. Do not mention the fact that you are an emigrant to persons who have no business to know it.

3. Never carry any large sum of money with you; you can always buy drafts at banks, and if you are going to a strange place you can give your photograph to the banker to forward to your destination, so that you can be identified without trouble when you want to draw your money.

4. Avoid those strangers who claim to be old acquaintances, and whom you do not recollect. A certain class of thieves claim the acquaintance of ignorant countrymen whom they want to rob.

5. Do not drink at the solicitation of strangers; the first point of the thief is to intoxicate or drug his victim.

6. Do not play cards for money with strangers; in many cases they confederate to rob emigrants.

7. Travel in company with old friends, if possible, and do not leave them. Thieves prefer to take their victims one at a time.

8. If you see anybody pick up a full pocket-book, and he offers it to you for a small sum; or if you see some men playing cards, and you are requested to bet on some point where it seems certain that you must win; or if you see an auctioneer selling a fine gold watch for five dollars, don't let them catch you. Emigrants are systematically swindled by such tricks.

9. If, when you arrive in a strange town, you want information and advice, you can always get it by applying at the right place. First, apply at the office of the Immigration Society, if there is one. If you are a foreigner, you will probably find in the large cities a Consular office or a benevolent society of your countrymen, and you can apply there. Usually, there are attentive and polite men at the police office. Public officers generally in the United States are ready to assist and advise strangers.

10. Before starting from home, carefully read all the accessible books about the State or Territory to which you intend to go; and when you arrive, go to some place where you can find old friends, if you have any. If you are poor, commence work immediately, but do not be in a hurry to buy land, unless with the approval of men whom you can trust. Take a month or two to get information about the country. Advice about the purchase of land is often given with corrupt motives.

WHITE MOUNTAIN RESERVATION.—The boundaries of the reservation to be as follows, as shown in red on the accompanying map: Starting at the point of intersection of the boundary between New Mexico and Arizona with the south edge of the Black Mesa, and following the southern edge of the Black Mesa to a point due north of Sombrero or Plumoso Butte; then due south to said Sombrero or Plumoso Butte; then in the direction of the Picache Colorado to the crest of the Apache Mountains, following said crest down the Salt River to Pinal Creek, and then up the Pinal Creek to the top of the Pinal Mountains; then following the crest of the Pinal range, “the Cordilleras de la Gila,” the “Almagra Mountains,” and other mountains bordering the north bank of the Gila River to the New Mexican boundary, near Steeple Rock; then following said boundary north to its intersection with the south edge of the Black Mesa, the starting point.

Southern Pacific Railroad.

Regular and Special Rates in U. S. Gold Coin for the “Loop Route.”

BETWEEN SAN FRANCISCO AND	NEWHALL.	SAN BUENA-VENTURA	SANTA BARBARA	LOS ANGELES	SAN DIEGO	COLTON	YUMA
Distances	438 M	488 M	518 M	470 M	606 M	528 M	720 M
1 Unlimited First Class	\$25.85	\$30.85	\$33.85	\$28.00	\$38.00	\$31.00	\$50.00
2 Unlimited First Class including Yosemite				65.00			
3 Limited First Class	20.00	24.00	27.00	20.00	30.00	23.00	42.00
4 Limited Third Class	10.00	15.00	18.00	10.00	20.00	13.00	32.00

[1](#) Allows stop-over privileges, at pleasure, upon notifying Conductors.

[2](#) Includes the Tourists' Trip from Merced to Yosemite and Return.

[3](#) Limited to a continuous trip between San Francisco and Los Angeles.

[4](#) Limited to a continuous trip between San Francisco and Los Angeles, on Third Class Trains.

T. H. GOODMAN,
General Pass. & Ticket Agent.

Eastern Railroad Rates.

FIRST-CLASS.

<i>From</i>	<i>Denver or Colorado Springs.</i>
St. Louis to	\$50.00
Chicago to	65.00
Cincinnati to	65.00
Quincy to	49.00

Kansas City to Atchison to	45.00 45.00
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EMIGRANT RATES.

<i>From</i>	<i>Denver and Pueblo.</i>	<i>La Veta.</i>
New York to	\$31.55	
St. Louis to	22.00	\$25.00
Cincinnati to	30.00	33.00
Quincy to	22.40	25.40
Chicago to	29.40	32.40
Toledo to	34.40	37.40
Indianapolis to	29.00	32.00
Kansas City to	} 20.00	24.00
Atchison to		

FREIGHT RATES.

Household goods, trees and shrubbery, farm implements, wagons, stock, old mining tools, etc., emigrant's account only, from Kansas City to Denver, Colorado Springs, and Pueblo, \$100 per car; less than car loads, \$1 per 100 lbs. To El Moro, \$130 per car; less than car loads, \$1.30 per 100 lbs.

FROM MISSOURI RIVER TO COLORADO, EN ROUTE TO ARIZONA.

First-Class Fare.—Atchison or Kansas City to Pueblo, Colorado Springs, and Denver, \$35; Veta, \$38; Cañon City, \$36.25.

Second-Class Fare.—From Kansas City, Atchison, Topeka, and Lawrence to Pueblo and Denver, \$30; Veta, \$33.75; Cañon City, \$32; El Moro, \$35; Del Norte, \$51.75; Lake City, \$67.75; Santa Fé, \$75.

From Kansas City, Atchison, Topeka, and Lawrence to West Las Animas, \$26.

Emigrant Rates.—From Kansas City, Atchison, Topeka, and Lawrence to West Las Animas, La Junta, Pueblo, and Denver, \$20; Veta, \$24; Cañon City, \$22.50; El Moro, \$25.50; Del Norte, \$42.50; Lake City, \$58; Santa Fé, \$65.50.

Distance to Santa Fé, New Mexico.—From Atchison or Kansas City to Trinidad, Col., 707 miles by rail; from Trinidad, by stage, to Cimarron, 74 miles; to Fort Union, 14 miles; to Las Végas, 139 miles; and to Santa Fé, 214 miles; making the total distance from Atchison or Kansas City 923 miles—707 miles by rail and 216 by stage. From Santa Fé to Mesilla, 180 miles by stage; to Silver City, from the same, 385 miles. At this point, connection is made by the Southern Pacific mail stages either for El Paso and the Texas routes, via Mesilla, or northward, by way of Santa Fé and as above, to the Colorado and Kansas railroads.

FARES FROM SAN FRANCISCO.

By Coast Steamer.—San Pedro or Santa Monica—cabin, \$14; steerage, \$9, including berth and meals; time about 45 hours to Santa Monica; leave San Francisco every two or three days, at 9 A. M. San Francisco to San Diego—cabin, \$15; steerage, \$10; San Pedro to Los Angeles, 50 cents; Santa Monica to Los Angeles, \$1.

Los Angeles to Yuma, \$23, railroad; Los Angeles to Dos Palmas, \$13.10, railroad; Dos Palmas to Ehrenberg, stage, \$20; Los Angeles to Colton, \$3; Colton to Yuma, \$19; Colton to Dos Palmas, \$10.10. Sleeping berths, (two nights) \$5. Yuma to Ehrenberg, steamer—cabin, \$15; deck, \$10. San Francisco to Ehrenberg, (by steamer, cabin, to Santa Monica; thence railroad to Dos Palmas, and stage to Ehrenberg) \$48.10. Same points, by railroad to Yuma and steamboat thence, \$65; or by railroad to Dos Palmas and stage thence, \$55.10.

Stage Fares.—Tucson to Altar, Sonora, \$10; to Hermosillo, Sonora, \$20. Leave Tucson every Monday noon; return Saturdays, 10 A. M.

Express Matter.—On 25 pounds and over, to Altar, four cents per pound; to Hermosillo, eight cents per pound.

The fare to and from Prescott to Los Angeles, San Francisco, Sacramento, or San José, by California and Arizona stage to Dos Palmas, and thence by rail, is as follows:

To Los Angeles	\$61.25
To San José	86.75
To San Francisco	86.75
To Sacramento	86.75

The person holding a ticket has the privilege of stopping over in Los Angeles or other points on the railroad for a reasonable length of time.

Schedule Time by Overland Stage from Yuma, leaving San Francisco by Southern Pacific R. R. thereto

<i>Time.</i>	<i>Arrives.</i>		<i>Time.</i>	<i>Arrives.</i>	
2nd day	1 P. M.	Gila City	Ar.	7th day	11 A. M. Silver City N. M.
3rd "	9 A. M.	STANWIX	"	8th "	12 M. MESILLA "
3rd "	7 P. M.	Gila Bend	"	8th "	1 P. M. Las Cruces "
4th "	5 A. M.	Maricopa Wells	"	9th "	1 A. M. EL PASO Tex.
4th "	10 A. M.	PHENIX	"	11th "	12 MN. Fort Davis "
4th "	3 P. M.	FLORENCE	"	12th "	1 A. M. Fort Stockton "
4th "	5 P. M.	Camp McDowell	"	13th "	2 A. M. Fort Concho "
5th "	6 A. M.	TUCSON	"	14th "	11 P. M. Fort Worth "
6th "	6 A. M.	Camp Grant	"	14th "	11 P. M. San Antonio "
6th "	11 A. M.	Camp Bowie	"	14th "	11 P. M. AUSTIN "

United States Military Telegraph—California, Arizona and New Mexico Divisions.

TABLE OF DISTANCES.

<i>Stations.</i>	<i>Miles.</i>
San Diego, Cal.	0
Campo, Cal.	43
Yuma, A. T.	173
Stanwix, A. T.	269
Maricopa Wells, A. T.	364
Florence, A. T.	410
Tucson, A. T.	473
Tres Alamos, A. T.	519
Camp Grant, A. T.	572
Camp Bowie, A. T.	617

PRESCOTT BRANCH.

Phoenix, A. T.	382
Wickenburg, A. T.	432
Prescott, A. T.	487
Camp Verde, A. T.	523

APACHE BRANCH.

Camp Grant	572
Camp Goodwin	612
Camp Apache	697
Ralston, New Mexico	663
Silver City, New Mexico	712
Fort Bayard, New Mexico	721
Fort Cummings, New Mexico	760
Mesilla, New Mexico	811
Los Cruces, New Mexico	813
Fort Selden, New Mexico	831
Fort McRae, New Mexico	880
Fort Craig, New Mexico	922
Albuquerque, New Mexico	1,033

Bernalillo, New Mexico	1,051
Santa Fé, New Mexico	1,097

ITINERARY.

STAGE, MILITARY, AND MINE ROADS, STATIONS, TOWNS, AND MILITARY
POSTS. WATER, WOOD, GRASS, ETC., EN ROUTE. COMPILED FROM THE
BEST MILITARY AND OTHER AUTHORITIES.

Yuma, A. T., to Mesilla, N. M.—Overland Road and Stations of the S. P. O. M. S. Line.

	<i>Miles.</i>	
YUMA, A. T.*		
Desconso	14	14
Gila City	8	22
Rattlesnake	7	29
Mission Camp ¹	3	32
Filibuster	12	44
Antelope Peak	6	50
Mohawk ²	16	66
Teamster's Camp ³	20	86
Stanwix*	20	96
Burke's ⁴	12	108
Oatman Flat ⁵	10	118
Gila Bend ⁶	30	148
Maricopa Wells* ⁷	43	191
Pima Villages ⁸	12	203
Sweet Water.	7	209
Sacaton.	6	215
Montezuma.	10	225
Sanford ⁹	8	233
Florence* ¹⁰	4	237
Desert Wells	36	273
Point of Mountain	9	282
Water Holes	9	291
TUCSON* ¹¹	9	300
Cienega ¹²	30	330
Tres Alimos ¹³	20	350
Steel's Ranch ¹⁴	40	390
Apache Pass* (Camp Bowie) ¹⁵	35	425
Ralston, New Mexico	50	475
Knight's Ranch, New Mexico	25	500
Silver City, " ¹⁶	25	525
Fort Bayard, "	10	535
Rio Membres, "	40	565

Fort Cummings*	”	20	585
Slocum’s,	”	30	615
MESILLA,	” 17	15	645

* Telegraph Offices.

At all stations, wood, water, and grass, or other feed, are to be found. At most of them, a small store is also kept. Meals and lodgings can be obtained.

[1](#) At Mission Camp, there is a road south and west, to the old Papago country, via Cabaza Prieto.

[2](#) From Mohawk is a road due south to the same region.

[3](#) Five miles east of Teamster Camp is a route south to the Ajo Copper Mines, and then southwest to Sonora.

[4](#) At Burke’s is a road south to the same point.

[5](#) At Painted Rock, north of and near this station, a road to Phoenix comes in.

[6](#) Near Gila Bend, a road diverges south, through the old Papago country, to old mines and deserted Indian villages.

[7](#) At Maricopa Wells, roads north and south diverge direct to Tucson, the Papagonia, to Phoenix, Camps McDowell, Verde, and Apache.

[8](#) At Pima Villages, a road diverges north, passing through the Maricopa village (Indians, 800). There are 5,000 Pima Indians, living in ten large and several small villages, between this station and Montezuma. There is also a road directly south from Pima.

[9](#) Road south to Desert Wells.

[10](#) From Florence, regular stages leave for Silver City, Pioneer District; Globe City, Globe District; San Carlos, and Camp Apache; also, to Hayden, East Phoenix, Phoenix, Wickenburg, and Prescott. There is a direct road to old Camp Grant, east, and thence to the San Pablo Valley and Arivipa settlements.

[11](#) At Tucson, the highway to Mexico, via the valley of the Santa Cruz, diverges west. Stages leave for Guaymas.

[12](#) Near Cienega, a road diverges to Davidson’s, the Santa Rita placers, old Camps Crittenden and Wallen, the Pategonias, and the old Presidio San Pedro (U. S. Camp Huachuca).

[13](#) At Tres Alimos, there is a road north and up the San Pedro Valley to San Carlos, the Globe District, and Camp Apache.

[14](#) There are roads north and south at this point to Camp Grant and the old Chiricahua Reservation, now abandoned, and thence, by trails, etc., to Sulphur Springs Valley, Dragoon Mountains, etc.

[15](#) There is a road north direct to Safford, on the Gila.

[16](#) Stages here for the most southerly station, via Santa Fé, on the Denver & Rio Grande N. G. R. R.; thence east, via Colorado and Kansas railroads.

[17](#) At Mesilla, the overland stages connect with branch to El Paso, Fort Worth, etc., Texas.

Distances Between Stations, commencing at Dos Palmas, on S. P. R. R., and thence over Lines of Cal. and Arizona Stage Co.

			<i>Miles.</i>
Dos Palmas	west to	Canyon Springs	15
Canyon Sp’gs	”	Chuckawalla	36 51
Chuckawalla	”	Mule Springs	18 69
Mule Springs	”	Willows	29 98
Willows	”	EHRENBERG 1	11 109
Ehrenberg	”	Tyson’s Wells 2	22 131
Tyson’s Wells	”	Desert Well 3	28 159
Desert Well	”	Mungia Well	17 176
Mungia Well	”	Cullings Well 4	15 191
Cullings Well	”	Point Mountain	25 216
Point Mountain	”	WICKENBURG 5	20 236
Wickenburg	north to	Partridge City	17 253
Partridge City	”	Antelope Valley	10 26

Antelope Val'y "	Dixon	17 280
Dixon "	PRESCOTT ⁶	17 297
Wickenburg south to	Smith's Mills ⁷	15
Smith's Mills "	Agua Fria	28 43
Agua Fria "	PHENIX ⁸	22 65
Phœnix "	Hayden Ferry	9 74
Hayden Ferry "	House's Well	20 94
House's Well "	Florence	20 114
Florence "	Tucson (overland stage)	63 277

- ¹ Ehrenberg is a steamboat landing of importance. Roads north to La Paz, the Colorado Reservation Camp and Agency, and south to Eureka and Castle Dome district.
- ² A road branches here southward to Castle Dome landing, and the mines in that district.
- ³ At this station mine roads diverge to the Harcuvar District and mines; an extensive stock range is found within five miles.
- ⁴ At Cullings, the road forks, the right-hand branch going to Wickenburg, and the other striking more northerly, via Date Creek, to Prescott.
- ⁵ At this point roads diverge south to Phœnix and Florence, to the Vulture Mine, to Agua Fria and Cave Creek mines, east and north to Prescott, and northeast to adjacent mining districts.
- ⁶ At Prescott, roads diverge to Camp Verde and valley, with its growing settlements, to Alexandria, and other mining towns and camps; north to the Black Hills and Forest, and to the San Francisco Mountain; northwest to Mineral Park and vicinity, and almost due west to Aubry and Hardyville landings. Also east, via Camp Verde, across the Plateau to Fort Wingate, New Mexico, and thence to the present southern terminus of the Denver and Rio Grande Railroad.
- ⁷ At Smith's Mill, south of Wickenburg, a road east diverges to Camp McDowell.
- ⁸ At Phœnix, roads diverge southward, one crossing to and following the Hassayampa Creek to the Gila, others eastward, up the Salt River valley, to Marysville and Camp McDowell, while other roads strike south, direct to Maricopa Wells, the mail stage line diverging in a southeast direction to Florence.

Military Routes.

NUMBERS 1-9 include all the Routes north of the Gila River; NUMBERS 14-25 include all the routes south of the Gila River in Arizona, and the Routes from San Diego to Fort Yuma; NUMBERS 26-31 include the Routes from Arizona into New Mexico and Sonora, from Fort Whipple, A. T., to Fort Wingate, New Mexico, and the Routes from Camp Pinal and Camp Apache.

☞ When there is more than one route, the distance on the usual route is marked with an Asterisk (*). There is a difference of ten per cent. less, as a rule, on the roads and routes surveyed by the military authorities, as compared with those given by the stage companies and mail contractors.

No.	Route.	Distance.
1	Camp Mojave to Willow Grove	79.78
2	Camp Willow Grove to Fort Whipple (PRESCOTT)	84.88
3	Fort Whipple to Camp Verde	38.59
4	Fort Whipple to Date Creek	59.65
5	Ehrenberg to Camp Colorado (Indian Reservation)	45.50
6	Ehrenberg to Date Creek	{ *130.32
		137.17
7	Ehrenberg to Wickenburg, A. T.	131.32

8	Camp Colorado to Date Creek	{	*175.82
			84.00
9	Date Creek to Camp McDowell	{	*109.83
			126.18
10	Date Creek to Maricopa Wells		101.05
11	Wickenburg to Fort Whipple, (PRESCOTT) via trail		55.00
12	Wickenburg to Camp McDowell, via trail		65.00
13	Camp McDowell to Fort Whipple (PRESCOTT)		109.85
14	Camp McDowell to Camp Reno		33.00
15	Camp Reno to Camp Verde		90.00
16	Camp McDowell to Maricopa Wells		44.81
17	Camp McDowell to Camp Grant		106.17
18	San Diego to Fort Yuma, Cal.	{	*191.61
			229.05
19	Fort Yuma, Cal., to Maricopa Wells (YUMA)		176.73
20	Maricopa Wells to Camp Grant		90.78
21	Camp Grant to Camp Goodwin		153.46
22	Camp Grant to Camp Bowie		135.34
23	Maricopa Wells to Tucson (Camp Lowell)		98.01
24	Camp Grant to TUCSON (Camp Lowell)		52.08
25	TUCSON (Camp Lowell) to Camp Goodwin		149.85
26	TUCSON (Camp Lowell) to Camp Bowie		105.36
27	TUCSON (Camp Lowell) to Camp Crittenden (east of Santa Rita)	{	*50.78
			87.36
28	Camp Crittenden to Camp Bowie		97.55
29	Camp Bowie to Camp Goodwin	{	*88.00
			99.00
30	Camp Bowie, A. T., to Fort Cummings		114.00
31	TUCSON (Camp Lowell) A. T., to Guaymas, Mexico		350.78
32	Camp Wallen, (abandoned) A. T., to Guaymas, Mexico		317.50
33	TUCSON, (Camp Lowell) A. T., to La Libertad, Mexico	{	*225.23
			226.97
34	TUCSON, (Camp Lowell) A. T., to Lobos, Mexico		213.64
35	Camp Pinal to Tucson (Camp Lowell)		114.63
36	Camp Pinal to Fort Whipple		243.97
37	Camp Pinal to Camp Grant		87.65
38	Camp Apache to Tucson (Camp Lowell)		221.85
39	Camp Apache to Maricopa Wells		316.24
40	Camp Apache to Fort Whipple		268.00

From Yuma.

<i>To</i>		<i>Miles.</i>	<i>Route</i>
Camp Pinal,	disused	260	Via Florence, on the Overland Road.
Camp Apache		497	Via both Tucson and via Camp Grant, 494 miles.
Camp Bowie		380	
Camp Colorado,	disused	195	Via Ehrenberg and then by river road, 45 miles.
Camp Colorado,	"	215	Via river steamer.
Camp Crittenden,	"	326	Via Maricopa Wells, Tucson and Davidson's Springs.
Date Creek,	"	278	Via Ehrenberg or via Camp Colorado.
Camp Goodwin,	"	425	
Camp Grant		268	Via Maricopa Wells and Florence or Tucson.
Tucson		275	Via direct from Maricopa, by stage road 300 miles.
Camp McDowell		222	Via overland road to Maricopa Wells.
Camp Mojave		503	By river steamer.
Camp Reno,	disused	255	Via Camp Verde.
Camp Verde		377	Via overland road to Maricopa Wells, and thence via Phoenix and Camp McDowell.
Prescott		338	Via Oatman Flat and Wickenburg.
Ehrenberg		140	By river steamer.
Maricopa Wells		177	By regular stage route, 191 miles.
La Paz		130	By river steamer.
Guaymas, Mexico		620	Via Tucson.
La Libertad, Mexico		500	Via Tucson.
Lobos, Mexico		489	
Tubac		321	Via Maricopa Wells direct to Tucson, regular stage route 346 ms.
Castle Dome Mills		12	
" Mines		22	
" Landing		22	By river steamer.
Sonora Line		50	" "
Eureka			" "
Landing			" "
Aubry			" "
Hardyville			" "
Callville			" "

From Prescott.

<i>To</i>	<i>Miles.</i>	<i>To</i>	<i>Miles.</i>
Camp Pinal, disused	244	Camp Toll Gate, disused	39
Camp Apache	481	Camp Verde	39
Camp Bowie	364	Fort Cummings, N. M.	478
Camp Colorado, disused	236	Fort Yuma, Cal. (Yuma, A. T.)	338
Camp Crittenden, ”	310	Ehrenberg, A. T.	190
Camp Date Creek, ”	60	Maricopa Wells	161
Camp Goodwin, ”	409	Guaymas, Mexico	610
Camp Grant, A. T.	252	La Libertad, Mexico	484
Camp Lowell, (Tucson)	259	Lobos, Mexico	473
Camp McDowell	170	San Diego, Cal.	530
Camp Mojave	165	Tubac	305
Camp Reno, disused	203	Fort Wingate, N. M.	286

Camp Mojave to Willow Grove, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Hardyville	6	6	Village; sandy road.
Alexander's Camp	2	8	Water and wood; no grass.
First Water, Union Pass	11	20	Water; grass scarce; no wood; no camping ground.
Union Pass (Spring)	1	21	Water; no wood or grass; road up hill; no camping ground.
Coyote Spring	16	38	Water; grass scarce; wood.
Beale's Spring	1	39	Another spring ½ mile beyond; very good water.
Hualpais Spring	14	54	Half a mile to right of road; water bad; good grass.
Tanks	12	67	Filled with sand; no water or wood.
COTTONWOOD	7	75	Water, grass and wood.
WILLOW GROVE	4	81	With the fractions added.

1½ miles beyond Beale's Spring, where road crosses Sandy Wash, there is permanent and good water; ¼ mile to right of road, in the Wash, grass abundant; good camping ground. Wood near by, ¾ mile to left of road. Up the Wash are large bodies of good water.

3 miles before reaching Hualpais Spring, just after crossing Big Wash, good permanent water; grass and wood 1½ miles to right of road; good road nearly to the water. In coming from Willow Grove to Camp Mojave, in order to reach this camping ground turn to left 2 miles after leaving Hualpais Spring. Important camping ground, used by trains.

Willow Grove to Prescott, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Fort Rock	9	9	Ranch. Water, grass and wood. Road generally good.
Camp near Muddy Cañon	11	20	Water in cañon 300 yards to right of road; wood abundant; road good.

Anvil Rock	4	24	Water and grass.
Oaks and Willows	9	33	Water, grass and wood. Road generally good.
Old Toll Gate	9	43	Abandoned. Road hilly, otherwise good.
Roblett's (Ranch)	2	45	Water, grass and wood.
TOLL GATE (Ranch)	} 1	46	Water and wood abundant. Road as above.
Camp Hualpai			
Williamson's Valley	15	62	Water and grass; no wood. Road excellent.
Lee's Ranch	13	75	Water, grass and wood. Road excellent.
PRESCOTT	11	86	Road excellent.

Three miles beyond Camp near Muddy Cañon is an old government camping ground, with water all the year; wood and grass abundant. New road forks to the left, one mile beyond Camp. Two miles further is the Camp opposite the above mentioned water, one-fourth mile to right of road. Present camping ground well marked.

This new road intersects old road one mile before reaching Anvil Rock; is smooth, and avoids the rocky hills on the old road, now very difficult for loaded teams. Both roads are boggy in winter.

Prescott (Fort Whipple) to Camp Verde, A. T. Route 1.

To	Miles.	Miles.	Description.
¹ Lurty's Ranch	11	11	Water and grass; wood scarce. Good road.
Ash Creek	9	21	Water permanent; grazing tolerably fair; wood scarce.
Cienega	7	28	Water permanent; grazing excellent; wood close by spring.
Summit Grief Hill	4	32	Water to right of road one mile before reaching Summit except in dry season; grazing good; wood plenty.
CAMP VERDE	5	38	Road good; first mile steep descent.

¹ The road forking to the right at this place leads to Agua Fria Ranch, distance 4 miles; from Agua Fria Ranch to Ash Creek by a direct road the distance is 7.80 miles.
The direct road from Lurty's cannot be used for supply trains, which all go by Bower's Ranch, making the distance from Fort Whipple to Camp Verde 40.67 miles.

Prescott, (Fort Whipple) to Camp Verde, A. T. Route 2.

To	Miles.	Miles.	Description.
Lurty's Ranch	14	14	Via "Point of Rocks." Water and grass; wood scarce; road good.
New Road to Camp McDowell	6	20	
Ash Creek	2	23	Water permanent; grazing fair; wood scarce.
Government Saw Mill	6	30	
CAMP VERDE	16	46	Good road.

Captain Foster, Assistant Quartermaster, reported this road in 1874 as the only practicable one for loaded wagons, between Fort Whipple and Camp Verde.

Prescott to Date Creek, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Lee's Ranch	11	11	Water, grass and wood. Right-hand road to Camp Mojave.
Tonto Spring	9	20.60	Water to right of road ¼ mile distant. Good camping ground.
Dickson's Ranch	10	30.60	Water, grass and wood. Road good.
Ehle's Ranch	3	34.10	Water, grass and grain. Road good. Mail station; good stabling.
Uncle Rob's	5	39.79	Water, grass and grain.
Jones' Camp	3	42.79	Water, part of year; grass good. Road bad and dangerous.
Willow Spring	4	47.71	Water ¼ mile to left of road; grass good; wood scarce.
Soldiers' Holes	6	53.71	Water in rainy season.
DATE CREEK	5	60.00	Road mostly good.

By a rough trail from Fort Whipple over Granite Mountains to Ehle's Ranch (Skull Valley) the distance is estimated at eighteen miles. There is another trail leading over this range entering Skull Valley at its upper end, (Dickson's Ranch) three and one-half miles from Ehle's. This trail is three or four miles longer than the other, but is not so rough. In winter the short trail (so-called) is often obstructed by snow. These trails are used by the "Mail Carrier."

Camp Apache to Prescott, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Tank	18	18	
Spring	5	23	
Forks of Road	8	31	Right fork to Zuni, N. M.
Jo. N.'s Camp	18	49	
Silver Spring	4	53	
Stoneman's Camp	8	61	
Shevelon's fork of Colorado	7	68	After crossing, take right fork of road.
Crossing of Little Colorado	28	96	
Sunset Crossing	33	129	
Tank	18	147	
Jo. N.'s Camp	20	167	
Sante Spring	8	175	
Stoneman's Lake	13	188	
Beaver Creek	18	206	
Camp Verde	18	224	
PRESCOTT	44	268	

Dos Palmas (S. P. R. R.) to Ehrenberg, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
¹ Dos Palmas (S. P. R. R.) Cañon Springs	11	190	Water and wood; little grass. Good road.
Chuc-a-walla (Station)	34	224	Water and wood. Good road.
Laguna	30	254	Water and wood; little grass; heavy sand.
Willow Springs	7	261	Water, grass and wood. Good road.
Bradshaw's Ferry (Ranch)	12	273	Water and wood. Good road.
² EHRENBURG	2	276	Sandy road.

¹ California and Arizona stages to Prescott leave this station.

² From Ehrenberg there is a road up the Colorado River to Camp Colorado, 45.50 miles.

Irrigated lands in Southern Arizona will readily produce two crops of grain each year, and several of alfalfa. Some lands, belonging to the Pima Indians, as well as fields about Tucson and the ranches of the Santa Cruz and San Pedro Valleys, are known to have been in continuous cultivation for at least from two to three hundred years. Water fertilizes and restores the soil. The valley of the Gila, under analysis, shows more phosphorates and other fertilizers than that of the Nile.

A sand storm on the mesas of Southern Arizona is not a pleasant affair to encounter. If caught in one on horse-back or afoot, imitate the animals, put your face close to the ground and turn your back to the blast.

Ehrenberg to Camp McDowell, A. T., via Date Creek.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Tyson's (Los Pasos)	25	25	Good water; hay and grain.
Desert Station	25	51	Good water; hay and grain.
Flint's	19	70	Good water and grass.
McMullen's	4	74	Old road branches here } Road by Martinez's Cañon. } west of
Cullen's	10	85	Good water and grass. } Date Creek
Date Creek Crossing	36	121	Good water and grass. } Mountain.
DATE CREEK	9	130	
Martinez's Cañon	7	7	Water, grass and wood; road boggy in wet season. Right fork to Ehrenberg.
Vulture Mill	18	26	Last six miles of road sandy.
¹ WICKENBURG	1	27	Settlement.
² Camp on Hassyampa	5	33	Water (except in very dry season), grass and wood.
Road leaves Hassyampa	2	35	Quicksands in Hassyampa sometimes impassable. (See Note.)
Mud Tanks	12	48	Water in rainy season.
Point of Mountain	7	55	Permanent water in White Tanks, 1½ miles to right of road.
³ Forks of Road		56	Right fork to Salinas Lower Crossing.
Crossing of Agua Frio Ranch	9	65	
Phoenix (Swelling's R'ch)	19	85	Settlement.
Acequia	1	86	

Forks of Road	12	98	Right fork to Maricopa Wells and Camp Grant.
CAMP McDOWELL	11	110	

- 1 The road runs along the bed of the stream for a part of the way; when the river is high quicksands are troublesome, and the road is sometimes impassable; whenever this is the case, the route is via the Vulture Mine, by which the distance is increased 18 miles.
From Wickenburg to Prescott, via Walnut Grove, the distance, by a very rough trail, is estimated at 55 miles. Ranch at Walnut Grove, half way.
From Wickenburg to Camp McDowell direct, by trail, the distance is estimated at 65 miles.
- 2 There is a road from this point down the Hassayampa to Burke's Station on the road between Fort Yuma and Maricopa Wells, with the following Camps:—Gila Bend, 40 miles; Cottonwoods, 25 miles; Camp opposite Oatman's Flat, 10 miles; Agua Caliente, 16 miles; Burke's Station (fording Gila River), 5 miles—total, 96 miles. The road is quite good, (though seldom traveled) and water, grass and wood are to be found at all the above Camps.
From Date Creek to Maricopa Wells, the road is taken to Salinas Lower Crossing (78 miles), thence by the Gila Lower Crossing (17 miles) to Maricopa Wells (6 miles)—total, 102 miles. In winter, when the Salinas and Gila are too high to ford, by going about 3 miles further up the Gila to Morgan's, advantage can be taken of a Ferry without increasing the distance to Maricopa Wells.
From Camp McDowell to Prescott, by a proposed wagon road that is opened from Prescott to Agua Frio, the distances are estimated as follows:—Camp McDowell to Agua Frio, 47½ miles; Dickson's Ranch, by newly opened road, 23 miles; Agua Frio Ranch, by old road, 18 miles; Prescott, 21 miles—total distance, 110 miles. The road from Camp McDowell to Camp Reno has the following camping grounds:—Camp Miller, 16 miles; Camp Carroll, 4½ miles; Camp O'Connell, 4 miles; Camp Reno, 8½ miles—distance, 33 miles.
From Camp Reno to Camp Verde, by a newly opened road, it is to Camp in Green Valley, 30 miles; thence to Camp Verde, by trail, 60 miles—total distance, 90 miles.
- 3 In rainy seasons, when the route by Agua Frio is impassable, it is necessary to take the right fork to Salinas Lower Crossing, (22 miles) thence up the north bank of the Salinas to Phoenix (about 23 miles)—total, 45 miles,—increasing the distance from Camp Date Creek to Camp McDowell about 16 miles.

Camp McDowell to Maricopa Wells, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Forks of Road	11	11	Right fork to Phoenix.
Ferry Station	2	13	Left fork to Camp Grant. Water, grass and wood. Crossing of Salinas River.
Desert Station	11	24	Well of water. Hay and grain at Station.
Morgan's Ferry	17	41	Crossing of Gila River.
MARICOPA WELLS	3	45	Stores. No grass or wood.

During the winter the Gila is usually and the Salinas occasionally, unfordable.

Camp McDowell to Camp Grant, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Forks of Road	11	11	Road excellent. Right fork to Phoenix.
Ferry Station	2	13	Small station; bad ford at high water.
Florence (crossing Gila)	38	52	Good fording; Ranch ½ mile this side.
Ruggles and Ewing	3	56	Last Ranch before leaving river; good stopping place.
Junction with Sacaton R'd	4	60	Desert mesa.
Round Valley	12	73	No water, wood or grass.
Camp near Round Valley	2	75	Water ½ mile to left of road by trail.

Cottonwoods	13	89	Water ½ mile to right of road by trail; grass; wood scarce.
Junction with Tucson R'd	13	103	
CAMP GRANT	3	106	Crossing Rio San Pedro.

At Prescott, clerks receive from \$50 to \$125 per month, with board often thrown in; carpenters and painters, from \$4 to \$6 per day; masons, from \$6 to \$8, and in some cases, when a man is possessed of superior skill, as high as \$10 per day; ranch hands, herders, cow-boys, from \$25 to \$50 per month, and board; common laborers, from \$2 to \$3 per day; domestic servants, men and women, from \$25 to \$40 per month; but as yet there is no great demand.

Maricopa Wells to Camp Grant, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Pima Villages	10	10	Store and mill.
Sweet Water	6	16	Store.
Sacaton	6	22	Store; water; grass scarce; right fork of road direct to Tucson.
Reservation, eastern boundary	7	29	
Walker's Ranch	6	35	Indian village and store.
White's Ranch	4	39	Gila; wood, hay, grain; little grass.
Junction with Camp McDowell Road	7	46	
CAMP GRANT	46	92	Crossing San Pedro.

The road from Maricopa Wells to Pima Villages is cut up with small gullies, from 1 to 4 feet deep, with steep sides, which, in rainy seasons, are muddy and troublesome.

The left fork leads up the Gila to Adamsville, 2½ miles distant, where are two stores, a mill, etc., and thence to Ruggles and Ewing's Ranch, (4 miles) where is a store; here the road intersects the road between Camps McDowell and Grant.

Camp Grant to Camp Goodwin.—In very rainy seasons it is necessary to go via Tucson, distance 202 miles. The shorter and better route, except in winter, is up the San Pedro River, 57 miles, to within 8 miles of Tres Alamos, where the left fork leads to Croton Springs, distance 25 miles, and thence to Camp Goodwin, 71 miles—total distance, 153 miles. On this road there are plenty of water, grass and wood, all along the San Pedro River.

Camp Grant to Camp Bowie.—To Croton Spring, distance 82 miles; thence to intersection with road between Tucson and Camp Bowie, distance 16 miles, and thence to Camp Bowie, 37 miles—total distance, 135 miles.

Maricopa Wells.

<i>To</i>	<i>Miles.</i>	<i>Total Miles.</i>
Yuma		191
Tucson, southeast, (overland stage road)		109
Sacaton (en route direct to Tucson)		22
Blue Water " "	20	43

Picacho	13	57
Point of Mountains	24	81
Tucson	17	98
Camp Grant		90

This is a stage station, with stores, etc., of importance. It is the point of divergence for branch stages to Phoenix, Camp McDowell, and Camp Verde.

Fuller, in his Treatise on Silver Mines, says: "Wherever, in any part of the world, silver mines have been worked they are worked now, unless closed for war, invasion of Indians, etc. We know of no silver mines in the world that have given out." In support of this position, he instances the mines of Mexico, the old Spanish mines, (opened before Humboldt's time) the South American mines, still as productive as they were three centuries ago, mines in Hungary worked before the Christian era, the silver mines of Freiburg, opened in the 11th century, etc., nearly all now worked with unabated productiveness.

Maricopa Wells to Tucson, A. T.

(Going south direct.)

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Sacaton	22	22	Left fork of road to Camp Grant.
Blue Water	20	43	Well; grass and wood plenty; station; hay and grain.
Picacho	13	57	Grass and wood plenty; no water.
Mud Tanks	15	72	Water in wet weather, wood scarce.
Point of Mountain	8	81	Wells; grass plenty, wood scarce;
Nine Mile Water	8	89	station, hay and grain.
TUCSON	8	98	Capital of Territory. Road good after passing Pima Villages.

Camp Grant to Tucson, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
CAMP GRANT			Crossing of San Pedro
Forks of Road	2	2	Wood scarce; grass Right fork of road to Maricopa Wells.
Cañon del Oro	21	24	Water, grass, and wood plenty.
Water	5	30	Water scarce; grass and wood plenty.
Dry Camp	8	38	Water in wet weather; grass and wood plenty.
Roieta	8	47	Water in wet weather; grass and
TUCSON (CAMP LOWELL)	4	52	wood plenty.

The Rio San Pedro is sometimes impassable in winter on account of high water. The first nine miles of the road is in a cañon, level, and very sandy; the rest of the road to Cañon del Oro is hilly, ascending till near the cañon, when there is a long, steep descent. Three miles beyond Cañon del Oro the road enters the bed of a stream, usually dry; and continues in it to within a half mile of Dry Camp. At the foot of the mountains, opposite Dry Camp, say one and a half miles distant, are the ruins of an old Pueblo, where there is water all the year. The Roieta in winter is a running stream.

Tucson

<i>To</i>	<i>Miles.</i>	<i>To</i>	<i>Miles.</i>
Camp Pinal, disused	115	Camp Verde	298
Camp Apache	222	Fort Cummings, N. M.	219
Camp Bowie	165	Fort Whipple, (Prescott)	259
Camp Colorado, disused	349	Fort Yuma, Cal	275
Camp Crittenden, ”	51	Ehrenberg	303
Date Creek, ”	199	Guaymas, Mexico	351
Camp Goodwin, ”	150	La Libertad ”	225
Camp Grant	52	Maricopa Wells	98
Canip McDowell	143	Lobos, Mexico	214
Camp Mojave	424	San Diego, Cal	467
Camp Reno, disused	176	Tubac	46
Camp Toll-Gate ”	208		

Tucson to Camp Goodwin, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Forks of Road	14	14	Right fork to Camps Crittenden and Wallen.
Cienega (begins)	8	23	Water and wood plenty; grass scarce. Picket post.
Mescal Ranch	6	29	
Cienega (ends)		30	
Water Hole	8	38	Water in winter; grass plenty; wood scarce. Road forks to left to Tres Alamos.
¹ Crossing San Pedro	12	50	Water and grass; wood scarce. Picket post.
Forks of Road to Dragoon Springs	11	62	Water plenty at spring; grass and wood plenty. Right fork to Dragoon Springs, five miles distant.
Forks of Road to Camp Bowie	3	65	Grass plenty; wood scarce; no water. Right fork to Camp Bowie.
Croton Springs	13	78	Water brackish; grass plenty; wood scarce.
Oak Grove	16	94	Springs: grass and wood plenty.
Kennedy's Wells	3	97	Water poor; grass and wood plenty.
² Arivapa Creek	15	113	Water, except in very dry season; grass and wood plenty.
Eureka Springs	1	114	Grass; wood scarce.
Spring	8	122	
³ COTTONWOODS	8	131	Stream of water; grass and wood plenty. Road hilly.
CAMP GOODWIN	9	140	Road sandy and down hill.

¹ The road from Tucson is over a level mesa till it descends into a cañon, where the Cienega begins. There are several steep hills in the next few miles.

The banks of the San Pedro are high and steep, and about ten yards apart.

² In winter it is necessary to take the right fork to avoid Eureka Springs and the Cienega, just beyond it, which are then impassable. This road joins the one by Eureka Springs about a mile beyond the spring.

³ In summer the creek is dry at this point, but water can always be found by descending the creek half a mile.

Tucson to Camp Bowie, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Fork to Camp Goodwin	65	65	Left fork to Camp Goodwin.
¹ Junction of Road from Camp Crittenden	3	68	
Sulphur Springs	12	80	Water brackish; grass and wood scarce.
CAMP BOWIE	24	105	

¹ Half a mile further the road forks to the left, to Camp Goodwin.

The grasses in Arizona nearly all come up from the root, unlike those of California, which grow from the seed. Therefore, in Arizona, if there should be a year without rain, stock would not die of starvation. The nutritious gramma grass does not appear to run to seed at all.

Tucson to Camp Crittenden, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Forks of Road	14	14	Left fork to Camp Bowie.
Davidson's Spring	12	26	
Camp near Davidson's Spring	1	28	
Mescal Ranch	10	39	Left fork to Camp Wallen, (abandoned) distance twenty and a half miles. Fine country, grazing, water, and timber in abundance.
Road to Cienegas	1	40	
Junction of road from Wallen	7	48	
CAMP CRITTENDEN	2	50	

Tucson (via Tubac) to Camp Crittenden.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
San Xavier del Bac	8	8	Settlement of Papagos Indians. Old mission church.
La Punta de Agua	2	11	Ranch.
Saurita	9	20	Ranch.
¹ Canoé	12	32	No water in dry season; grass and wood plenty.
Tubac	13	45	Town. Point of departure for Sopori, Arivaca, Toltec Camps, Aztec District, for Santa Rita Mountains, etc.
Calabasas	12	58	Old Fort Mason.
² Smith's Ranch	3	61	

Sonoita	12	74	Vail's Ranch.
Old Fort Buchanan	12	86	Water, grass, and wood plenty.
CAMP CRITTENDEN	1	87	

- ¹ The left fork crosses the Santa Cruz at Canoé, recrossing the river near Tubac; (the measurement was made on this road). The right fork does not cross the river, and is longer.
- ² Just beyond Smith's Ranch, take the left fork to Camp Crittenden. The main road goes into Sonora. Between Tubac and Smith's Ranch, there are ranches every few miles, with water, grass, and wood. Between Sonoita and Camp Crittenden, there are several ranches with water and grass, but little wood. Road to Tubac very hard and smooth; but very rough from there to Crittenden, through Sonoita Cañon.

Camp Crittenden to Camp Bowie, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Forks of Road	1	1	Left fork to Tucson.
Forks of Road	13	14	Right fork to Santa Cruz.
CAMP WALLEN(abandoned)	5	20	On Babacomori Creek.
¹ San Pedro Crossing	18	38	Station.
Dragoon Springs	18	56	Water, grass, and wood.
² Junction with Road	3	60	Road from Tucson.
Sulphur Springs	12	72	Water brackish; grass and wood scarce.
CAMP BOWIE	25	98	

- ¹ Left fork leads down the right bank to the middle crossing of the San Pedro River, distance 13.32 miles, thence to Camp Bowie or Tucson.
- ² From this point to Camp Goodwin.

Camp Bowie to Camp Goodwin, A. T.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
¹ Forks of Road	6	6	Water usually: good grass. Left fork to Tank, half mile distant.
Water Holes	30	36	Water usually; good grass, near forks, on right side of road.
First Camp on Gila River.	27	63	Water; grass scarce.
Second Camp on Gila River	11	74	Water; grass scarce.
CAMP GOODWIN	14	88	

- ¹ The Tank always contains water. The road from the Tank to the Gila is over a grassy plain with no water in dry season, excepting at Water Holes, near the Junction with the old road along the Rio de Sauz. There water is usually found on the right-hand side of the road. There is another road to Camp Goodwin, via San Simon, (seventeen and a half miles) and thence down the Rio de Sauz, joining the above road at Water Holes, (thirty-one miles) making the distance about eleven miles longer.

Tucson, A. T., to Guaymas, Mexico.

[ESTIMATED.]

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
Smith's Ranch	61	61	
Los Nogales	5	66	
Agua Zarca	15	81	
La Casita	14	96	
Los Alisos	8	104	Water and grass; mesquite wood.
Imuris	11	115	
La Magdalena	11	127	
Santa Ana	12	139	
Bajorito	16	155	Wells and grass; mesquite wood.
Rancho Querobabi	23	178	
Rancho Tabique	28	206	Tanks and grass; mesquite wood.
Hacienda de Torreon	26	232	
Hacienda de La Labor	2	234	Water; grass scarce; mesquite wood.
Hacienda del Alamito	9	243	Water; grass plenty; mesquite wood.
HERMOSILLO	12	255	
Rancho de la Parza	16	271	Water; no grass; mesquite wood.
Rancho de la Palma	16	288	
Rancho del Posito	8	296	Tank; grass scarce; mesquite wood.
Rancho de lo Cienquito	15	312	
Rancho de la Mucho Bueno	19	331	No water; grass plenty; mesquite wood.
Rancho de la Caballo	9	340	Tank; grass plenty; mesquite wood.
GUAYMAS	11	351	

The road from Tucson to Guaymas, except 15 miles south of Calabasas, where it is heavy in wet weather, is one of the finest on the Pacific coast.

Tucson, A. T., to Port La Libertad, Mexico.

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
San Xavier del Bac	8	8	Settlement of Papago Indians.
La Punta de Agua	2	11	Ranch.
Sahuarito, (Columbus)	8	19	Water, grass and wood. Good road.
Roade's Ranch	8	28	
Los Taraises	2	31	
Reventon, (Kitchen's Ranch)	2	34	
Soporio Rancho	5	39	
Mina Colorado	11	51	
ARIVACA	7	58	

Los Alamos (Old Arivaca)	1	59	
Covodepe Cuesta (Mexico)	6	65	
Spring in bed of Arroyo	5	70	Water scarce; wood. Good road.
Z'Azabe	8	79	Water, grass and wood. Good road.
Charco de los Mesquites	6	86	No water.
Tecalote Trail		86	
Charco	4	90	Water, grass and wood.
Rancheria	2	92	Good grass.
Forks of Road	1	93	
Forks of Road		93	
Ascent to Mesa	1	94	
Tinaja, (Charco)	6	101	Water, grass and wood.
Los Paredones	15	116	Water, grass and wood. Good road.
Jesus Maria	14	130	Water and wood; grass scarce. Good road.
ALTAR	8	139	Water and wood; grass scarce. Good road.
Dry Arroyo	4	143	Good road.
Road to Zepedas Ranch	6	149	Good road.
Foot of Hill	2	151	Good hard road.
Summit of Hill	1	152	
PITIQUITO	1	153	Water, grass and wood. Good hard road.
Cienega and Caborca Road	7	161	Good hard road.
Laguna Mosca	5	166	No water in dry season; good grass, good hard road.
Bajia de Aquituna	6	172	Wood and grass; no water. Good level road.
El Zanjon (dry arroyo)	2	175	Wood and grass. Good road.
Tinaje del Viejo	7	182	Water.
Angostura Pass	7	190	Water, grass and wood. Good hard road.
Picu	11	201	Water, grass and wood. Good hard road.
Pozo de los Cristolas		201	
Charco de los Papagos	1	202	
Tinaja del Tule	4	206	Very little water or grass.
Derisadero Prieto	4	211	Wood; no grass. Good hard road.
Point where Gulf is first seen	1	212	
PORT OF LA LIBERTAD	13	226	Bad road.

Heavy blankets are a necessity in Arizona; the nights are always cool, even in the height of the "heated term." Woolen undergarments are desirable at all times.

Fort Wingate, N. M., to Prescott, A. T., via Camp Verde, A. T.

	<i>Miles</i>	<i>Yards</i>	<i>Miles</i>	<i>Yards</i>	<i>Description.</i>
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Ft. WINGATE, Spring Cr'k Crossing	2	1010	2	1010	Crossed by bridge, water plenty, wood on hills.
Spring	7	738	9	1748	Spring close to road on south side, at base rocky bluff, water bad, wood plenty, and good grazing, road sandy for short distance.
Bridge over Defiance road	3	496	13	484	Bridge across Rio Puerco of the West; water good and grass plenty.
Camp on Rio Puerco	3	525	16	1009	Water muddy, plenty wood, good grass.
Quirina Cañon	19	323	35	1332	Rio Puerco almost washing away the road. Bluffs on left bank very steep and abrupt.
Camp on Rio Puerco	5	442	41	14	Camp a short distance off road.
Crossing of Rio Puerco	12	659	53	673	Half a mile east of the crossing, a road leads off to the left, bed of river, quicksand, crossing fair, road good.
2d Crossing of Rio Puerco	1	540	54	1213	Here we were unable to effect a crossing, owing to recent freshet washing away the banks, leaving them 20 feet high and abrupt; left road and crossed country keeping from one to two miles from right bank of river. Road between crossings is sandy. The route on north side of Puerco is shortest. Distance to Carrizo Creek about 11 miles. Water in Rio Puerco, between those points, not permanent.
Camp on Rio Puerco	7	1345	62	798	Heavy traveling, water muddy, grass ordinarily good, greasewood abundant.
Road	13	834	75	1632	Traveling a little heavy; crossed some sandy Arroyos before getting on road, water in Rio Puerco.
Camp on Carrizo Creek	5	31	80	1663	Water obtained by digging, wood scarce, road and grass good.
Lithodendron Creek	13	1666	94	1569	A wide sandy bed, no water, crossing in dry weather good, but very difficult when there is water, quicksands, road good, a steep hill on east side. South of the crossing, the Rio Puerco becomes a wide, dry, sandy bed.
Camp on Little Colorado River	16	1216	111	1025	Water and grass abundant and good, plenty of wood, road good. About five miles up the Puerco, water was found in holes.
Camp on Little Colorado River	19	1020	131	285	Half a mile from road to river, plenty wood and water, grass good, road good.
Camp at Sunset Crossing Little Colorado River	14	493	145	778	Plenty wood and water, no grass near crossing, road good to Cottonwood Fork, which empties into the Little Colorado. near Sunset Crossing. Cottonwood Fork has a delta, and in time of freshet overflows the valley for several miles, rendering it impassable. Sunset Crossing is not passable in time of melting snows without the aid of a raft.
Camp at Sunset Pass, on Big Dry Fork	18	669	163	1447	Road gradually up grade, but good traveling, permanent water in tanks in bed of creek for about four miles, in cañon plenty wood, water, and grass.

Camp on a lake of snow water	21	42	184	1489	Four miles from Camp on Big Dry Fork, the road runs through thick cedar to Jarvis Pass, which is 14 miles from Sunset Pass. Road good to Jarvis Pass, thence stony; plenty of cedar on lake.
Simpkins' Spring	7	1681	192	411	Spring 100 yards north of road, (trees blazed) good water, grass ordinarily good, thick heavy pine, road stony and up grade.
Stoneman's Lake	10	859	203	510	Is about 4½ miles in circumference, circular, an abundance of permanent water. Lake inclosed by bluffs about 400 feet high, thick heavy pine, good grass, very difficult to get water. The road leads through the Mogollon Mountains from Simpkins' Spring to Stoneman's Lake. In the spring of the year, the road through the mountains is perfectly saturated with water, very miry, and impassable for heavy-loaded wagons. Pine timber is thick and heavy on mountains.
Bartlet's Tank	7	666	210	1173	Tank 400 yards north of road. Four miles from Stoneman's Lake, the road leads through thick cedar, and becomes very rocky. Two and a half miles southwest of lake are two small creeks with wood and grass, but no permanent water.
Beaver Creek Crossing	11	272	221	1445	The descent to the creek is very steep and abrupt. At base of hill, a trail leads southward to Camp Verde, which cuts off about 9 miles. Road up to this point leads through thick cedar, and is very rocky, thence good. Beaver Creek is a large stream of permanent water, rocky bed, banks low, crossing good, grass fair, plenty wood.
Rio Verde.	12	1121	234	806	Road good for 10 miles, thence hilly to crossing. Rio Verde 80 feet wide, gravel bed, good water, banks low, crossing good, scattered cottonwood on banks. A road leads up the left bank to the Indian Reservation. On right bank, a right-hand road direct to Prescott.
Camp Verde	6	1673	241	719	Road good.
Wild Cherry Creek	12	1412	254	371	Returned on road 4 miles, thence over foothills of Verde Mountains for 5 miles, thence ascend and descend mountains to Wild Cherry Creek. The ascent is very steep, but gradual; the descent is more abrupt; road in good order. Wild Cherry Creek is a running stream of permanent water. Timber and grass abundant.
Gayetty's Ranch	3		257	371	Ranch and station on left of road.
Ash Creek	6	233	263	604	Permanent water in holes south of crossing, banks low, rocky bed, crossing good, plenty wood and grass, road a little hilly. Three miles east is the junction of Grief Hill Road.
1st Crossing of Lynx Creek	7	1345	271	189	Ranch at crossing, creek dry, water in well, good grass, wood at ranch.
2d Crossing of Lynx Creek	8	1002	279	1191	Permanent water, ranch on west bank, scattered cottonwood, good grass, low banks, rocky bed, good crossing. Steep hill on either side.
FT. WHIPPLE	6	155	285	1346	On right bank of Granite Creek, road hilly.

Roads and Distances from the Colorado River (en route from Utah) South to Prescott.

NO. 1. FROM COLORADO CROSSING, VIA TRUXTON SPRINGS.

<i>To</i>	<i>Miles.</i>	<i>Total Miles.</i>	<i>Altitude.</i>	<i>Description.</i>
Tinnahkah Springs	21	21	4080.0	Small springs; bunch-grass; wood.
Attoovah (or Cañon) Springs	14	35		Spring in cañon; bunch-grass; cedar trees.
New Creek of Ives, or Pahroach Springs	14	49		Good camping-grounds; plenty wood, water and grass.
Truxton Springs	16	65	3885.5	Bunch-grass through sagebrush; water and wood.
Old Camp Willow Grove	25	90		Wood, water and grass;
Fort Rock	15	105		Good water; no grass; little wood.
Oaks and Willows	27	132		Water, wood and grass.
Old Camp Hualapais	9	141	5321.9	Good water and grass; plenty wood.
Toll-gate in Williamson's Valley	16	157		Water and wood; little grass.
PRESCOTT	23	180	5318.0	Water and grass.

Roads and Distances from the Colorado River South to Prescott.

NO. II. FROM MOUTH OF RIO VIRGEN, VIA VIRGIN AND BEALE'S SPRINGS.

<i>To</i>	<i>Miles.</i>	<i>Total Miles.</i>	<i>Altitude.</i>	<i>Description.</i>
Mountain Spring	41	41	5500.8	Water alkaline; little bunch-grass; wood.
Chloride City	14	55		Water brackish; little grass.
Mineral Park	7	62		Water alkaline; wood and grass at small distance from town.
Cerbat	6	68		Water and wood; very little grass.
Beale's Springs	9	77		Water and wood; grass some distance from camp.
Hualapais Springs	16	93		Good water, wood and grass.
Old Camp Willow Grove (Cottonwood)	20	113	4170.0	Good water, wood and grass.
Fort Rock	15	128		Good water; no grass; little wood.
Camp Hualapais	36	164	5321.9	Good water, wood and grass.
Toll-gate	16	180		Good water and wood; little grass.
PRESCOTT	23	203	5318.0	Good water and grass.

No III. From Moqui-Pueblos Trail, via Mouth of Paria Creek.

<i>To</i>	<i>Miles.</i>	<i>Total Miles.</i>	<i>Altitude.</i>	<i>Description.</i>
Moen-copie Creek	11	11	4984.1	From map; distance probably too small.

Colorado Chiquito	12	23		From map; distance probably too small.
Cascades	58	81		From map; water alkaline; wood; grass scarce on lava <i>débris</i> a few miles south of river.
Wagon road	11	92		Plenty of wood and grass.
Cosnino Tanks	4	96	6244.1	Wood and excellent grass; water said to exist in tanks all the year.
Antelope Springs	24	120	8065.1	Good grass and wood anywhere.
Volunteer Spring	11	133	7106.4	Good wood, water and grass.
Spring south of Bill Williams' Mountain	27	160	5526.6	Good wood, water and grass.
Rattlesnake Cañon	15	175	4600.0	Wood scarce; good water; bunch-grass.
Postal's Ranch	14	189		Good water; wood and grass poor.
PRESCOTT	22	211	5318.0	Plenty water and wood.

Road from Virgin to Mountain Spring generally good.

From Moqui-Pueblos trail to Cascades and from Cosnino Tanks distances were taken from map, and for road distances (though correction was made) are probably too small.

The trail used from Cascades on Colorado Chiquito to wagon road is perfectly practicable for wagon; hence good mail-road to Prescott.

Good camping ground at crossing of Muddy Cañon, between Fort Rock and Oaks and Willows.

Road from Mineral Park good.

Road from Cerbat sandy.

Road from Beale's Spring good; abandoned military post.

Road from Cottonwoods good. Mineral Park and Cerbat are both situated about one mile east of road from Chloride City to Beale's Spring.

To Navajo Springs, good made road round springs; excellent to Limestone Pockets and beyond, till it passes divide, when it becomes sandy.

From Navajo Springs to Moqui-Pueblos trail, good road, following arroyo.

Camp Wallen, (abandoned) A. T., to Guaymas, Mexico.

[ESTIMATED.]

<i>To</i>	<i>Miles.</i>	<i>Miles.</i>	<i>Description.</i>
CAMP WALLEN			Babocomori Ranch and Settlement.
Mescal Ranch	9	9	Water, grass and wood plenty.
San Rafael	8	27	Water and grass plenty; no wood.
Santa Cruz	8	35	Town; no wood.
San Lazaro	9	41	Water, grass and wood plenty.
Spring of water	12	56	
Cocospera	2	58	
Mouth of Cañon	6	64	

Babasaqui	12	76	Ranch.
Imeritz	3	79	Town.
Ternate	6	85	Flour mill. Water, grass and wood plenty.
La Magdalena	9	94	Town.
GUAYMAS	223	317	

Tucson

<i>To</i>	<i>Miles.</i>
Yuma (W by N)	300
Florence (N)	63
Silver City (E)	118
Globe City (N)	138
San Carlos (N by E)	175
Phœnix (N by W)	125
Copper Mines, Young America, etc. (W)	50
Maricopa Wells, (direct N by W)	98
Tubac (S)	46
Tumacacori (S)	49
Sonora line (S)	75
Cabajas (S)	
Arivaca (S by W)	
Hacienda del Santa Rita, Tyndall (S by E)	59
Salero House, Santa Rita (S by E)	60
Toltec C'mp, Aztec dist (S by E)	65
Sonoita, mill-sites, (S by E)	70
Oro Blanco, Ostrich mine, etc. (SW)	85
Tres Alamos (E)	50
Davidson Spring (SE)	
Old Camp Crittenden (SE)	
Babocomori Ranch, Camp Wallen (SE)	
Pategonia Mt'ns, Mowry Mine (SE)	85
Camp Hauchachi, near Old Presido, San Pedro (SE)	
Pueblo Viejo (NE)	150
San Carlos (NE)	160

Tucson to Camp Crittenden (east slope of Santa Rita Range) via Davidson's.

	<i>Miles.</i>	<i>Total Miles.</i>
Forks of Road (left fork to Bowie)	14	

Davidson's Spring	12	26
Mescal Ranch (left fork to Camp Wallen and Babocomori)	13	39
Road to Cienega	2	41
CAMP CRITTENDEN	9	50

The Same, via Tubac.

	<i>Miles.</i>	<i>Total Miles.</i>
Tucson to San Xavier del Bac (Papago Indian Reservation)	9	
La Punta de Aqua Ranch	2	11
Saurita Ranch	9	20
Canoe (no water in dry season, grass and wood abundant)	12	32
TUBAC	13	45
Calabasas	13	58
Smith's Ranch (main road to Sonora; left fork to Crittenden)	3	61
Sonoita (Aztec and Tubac mill-sites, ranch, and saw-mill)	13	74
CAMP CRITTENDEN	13	87

Prescott

<i>To</i>	<i>Miles.</i>
Nephi, Utah, U. South'n R. R. (N)	500
Fort Wingate, N. M. (E)	286
Present Terminus of Denver & Rio Grande N. G. R. R. (E)	540
Clifton, Longfellow Copper Mines, via Mogollon Plateau (E). Estimated	320
Wickenburg (S)	82
Phoenix (S by E)	142
Florence (S by E)	192
Tucson (S by E)	267
Tubac (S by E)	313
Camp Bowie (E by S)	392
Ehrenberg (SW)	213
Silver King (E of S)	190
Chino Valley (N)	22
Aqua Fria Valley (E)	15
Camp Verde (E)	42
Montezuma Wells (N of E)	55
San Francisco Mt'n (NE)	85
" " Forest (NE)	65
Black Cañon (SE)	52

Alexandria, Peck Mill (S)	7
Mill on the Hassayampa (S)	10
Walnut Grove, mill, etc. (S)	17
Colorado Chiquito, Sunset Crossing, (N of E)	132
Moqui Pueblos (N of E). Est'd	180
Stoneman's Lake (N of E)	73

Wickenburg

<i>To</i>	<i>Miles.</i>
Vulture Mills (N)	1
Smith's (S)	15
Vulture Mine (SE)	11
Aqua Fria (S)	43
Lambley's (S)	8
Cave Creek Mines (N by E)	40
Camp McDowell (SE)	95

From Tubac.

(Principal point in the Santa Cruz Valley, Santa Rita mining region.)

<i>To</i>	<i>Miles.</i>	<i>Total Miles.</i>
Tucson	46	
Tumacacori Mission (King's)	3	3
Old Hacienda del Santa Rita	9	12
TOLTEC CAMP, Aztec District	9	12
Sonoita, Aztec and Tubac Mill-sites	5½	17½
” , via Smith's Ranch		29
San Xavier del Bac		36
Reventon Ranch		6
Sopori		10
Arivaca		17
Calabasas		15
Cerro—Colorado District		22
Patagonia Mountains (Mowry Mines)		48
Fresnal		65
Ajo Copper Mines		135
Aliza Pass (Baboquivera Peak). Estimated		30
Canabi (Old Papago Country) ”		60
Papago Ranch (Sonora)		90

” ” (A. T.) by way of the Mexican Papago Ranch	113
Cayote Springs	45
Cuijota	72
Cholla	90
Saguarza	96
Santa Rosa	75
Pirigua	92
Sonoita (Sonora, by way of Old Papago Ranch)	160
St. Domingo (on Sonora line)	175
Camp Crittenden	42
” Wallen (Babocomori Ranch)	63
Camp Bowie (via Camp Crittenden)	139
Florence (via Tucson)	108
Yuma (via Tucson)	345
SAN FRANCISCO (via Yuma, Stage & S. P. R. R.)	1,065
Phœnix (via Florence)	157
Wickenburg (via Phœnix)	222
Ehrenberg (via Wickenburg)	349
El Paso, Texas, (via Tucson)	445
Mesilla, New Mexico, (via Tucson)	394
St. Louis (via N. M. Stage and Ks. R. Rds.)	1,778
Austin, Texas, (via El Paso)	1,095
Mexican towns (via Santa Cruz Valley):	
Magdalena	51
Santa Cruz	54
Altar	95
Hermosillo	229
Lobos	309
Guaymas ¹	229

¹ Ports in Sonora, on the Gulf of California.

Ehrenberg to

	<i>Miles.</i>
Wickenburg (E)	131
Prescott (E)	82

Mineral Park, Mojave County.

<i>To</i>	<i>Miles.</i>
Hardyville, Colorado River (S by W)	35
Cerbat, village (S)	6
McCracken Mine, Owen Dist. (S.)	100
Greenwood, mills, village, etc. (E of S)	100
Hackberry Mine (E)	35
Haulapai Mt'ns (SE)	80
" Camp (E)	103
Williamson's Valley (E)	121
PRESCOTT (S. E)	141

Phoenix

<i>To</i>	<i>Miles.</i>
Wickenburg (N)	60
Florence (S)	50
Maricopa Wells (SW)	35
Camp McDowell (E)	35
East Phoenix (E)	4
Ruins, north of river (E)	4
Hayden (E by S)	9
Ruins, near La Tempe (E by S)	16
House's Well (S)	20
Marysville (E)	18
Mount McDowell (E by N)	20
Placers, Superstition Mt'ns (E by S)	40

Florence

<i>To</i>	<i>Miles.</i>
Silver King, mine and mills, Pioneer Dist. (NE)	35
Globe City (G. D.) Pinal Mt'ns (NE)	75
Wheatfield, mines and furnaces (NE)	87
Stonewall Jackson, McMillen's Camp (NE)	93
San Carlos (NE)	115
Sanford (W)	6
Casa Grande (SW)	12
Adamsville (W)	5
Pima Villages (W)	34
Maricopa Villages (W by N)	42

Maricopa Wells	46
Tucson (SW)	63
Phoenix (NW)	50
Wickenburg (NW)	110
Prescott (NW)	192
Mineral Park (NW)	333
Yuma (W)	237
Tucson (S)	163
Prescott (NW)	192
Camp Grant (E)	50
Silver City, N. M. (E)	288

Railroad and Stage to Prescott.

The cheaper route to Prescott is from San Francisco via Dos Palmas, (160 miles east of Los Angeles) and stage thence through or via Ehrenburg—Dos Palmas to Prescott being about 200 miles.

Express trains leave San Francisco daily at 4 P. M.—arrive at Dos Palmas 2:10 A. M. second night.

Third class trains leave San Francisco daily at 4:30 P. M.—arrive at Dos Palmas at 2:10 P. M. third night.

Stage leaves Dos Palmas immediately after arrival of the train, every other night.

There are two rates of fare to Prescott via Dos Palmas and Ehrenburg. First class, \$78.10; third class, \$67.10, coin.

The most comfortable, but more expensive route, is via rail to Yuma River steamer to Ehrenburg, and stage thence to Prescott. But this connection depends upon the running of the steamer up the river from Yuma, of which the departures are irregular. Trains leave San Francisco: first class, at 4 P. M.—arrive at Yuma second morning at 8:30 A. M.; third class, at 4:30 P. M.—arrive at Yuma third morning at 8:30 A. M.

Fare from San Francisco to Yuma: first class, \$45; third class, \$34.

Yuma to Ehrenburg: cabin, \$15; deck, \$10. Ehrenburg to Prescott, \$33.

Through tickets are not issued by this route.

A daily stage connection is made from Yuma via Phoenix, by which route through tickets are sold from San Francisco to Prescott, at first class rates, \$117; third class being \$106.

Miscellaneous Distances.

	<i>Miles.</i>
Colton (S. P. R. R.) Cal., to Fort Mojave, A. T.	135
San Francisco to Yuma (S. P. R. R.)	720
St. Louis to Prescott (estimated)	1,500
St. Louis to Tucson (estimated)	1,500
St. Louis to Florence (estimated)	1,500
Fort Defiance to Prescott (estimated)	240

Papagoria Distances.

[ESTIMATED]

Santa Rosa to Cojéta	12
Cojéta to the Gila (Pima Villages)	50
Pirigua to the Gila (Cotterell's Station)	55
Pirigua to the Sonora line	40
Pirigua to the Saucita (north)	25
Ajo Copper Mines to the Gila (Burke's)	45
Ajo Copper Mines to the Gila (Mohawk Station)	50
Sonora line, via the Cabezo Prieta, to the Gila	70
Sonora line, via the Tinajaalta, to Yuma	110

Local Distances in Yavapai County

Prescott to Chino Valley	20
" " Agua Fria Valley	18
" " Alexandra	40
" " San Francisco Mountain	85
" " Walnut Grove.	25
" " Tiger Lode and Mine	40
" " Black Cañon	60
Azltan Mill to Black Cañon	75
Peck Mine to "	100
Walnut Grove to Tiger Lode	20
" " Peck Mine	30
Camp Verde to Beaver Creek	12
" " Hassayampa	10
Clifton to Longfellow Copper Mine	7
" " Silver City, N. M.	80
" " Coronado, "	10

Local Distances in Mojave County

Summit Springs to Aubrey	50
Mineral Mine to The Needles	45
McCracken Mine to Aubrey (via Planet Mine)	30
McCracken Mine to Parker	50
Signal to McCracken Mine	8
Signal to Greenwood	4

Colorado River Distances.

Miles. *Total
Miles.*

Point Isabel (Gulf of California) to Yuma		175
Yuma to Castle Dome Landing	35	210
” Ehrenberg	125	335
” William’s Fork	214	424
” Mojave Cañon	232	442
” Aubrey	220	395
” Chemchuevis Landing	240	405
” Mojave	300	465
” Hardyville	312	477
” Cottonwood Island	342	507
” Callville	402	567
” Stone’s Ferry	465	640

Local Distances in Pinal County.

Florence to Picket Post	25
Globe City to Pinal Creek	18

Local Distances in Pima County.

Tucson to Arivipa Cañon (N. E.)	120
Tucson to Picacho Mine (W.)	75

The famous Turquoise Mine is in New Mexico, near the Arizona line. It comprises two enormous open quarries, perhaps 200 feet in depth at the deepest point, and covering an area of several acres. They must have been produced with great labor, since there are no traces anywhere of the use of tools or gunpowder. Tradition refers these workings to a period of greater antiquity than the Spanish occupation, and declares them to have been executed by the Aztec inhabitants of the regions who preceded the present Indian races. Stone hammers have been found in these quarries, but no tools of any metal. The trachyte is seamed and fissured throughout, at small intervals, and in every direction; and there is no reason to doubt that hammers, wedges, and levers would be quite sufficient to remove the solid masses. The turquoise occurs fully in the fissures, in the form of narrow seams and plates, rarely or never exceeding the fraction of an inch in thickness. The majority of the seams now exposed show the impure green variety, which is worthless. The Pueblo Indians, like civilized people, value the light-blue turquoise only.

Stage arrives at Florence every morning at 7 A. M. from Tucson, and leaves half an hour later for Yuma; stage from Yuma arrives every evening at 6 P. M., and leaves for Tucson half an hour later. Stage to Globe City leaves every Tuesday at 8 A. M., and arrives at 8 P. M. Saturday. Phoenix and Prescott stages leave every other day at 7 A. M., and arrive alternate days at 12 midnight. Stage for Silver King leaves every Monday, Wednesday and Friday at 7 A. M., and arrives every Tuesday, Thursday and Saturday at 4 P. M.

The military-geographical surveys, up to the fall of 1876, under Lieut. Wheeler, in central and western Arizona, have covered a total area of 17,954.6 square miles, or 11,490,944 acres. Of this total, it is estimated that 25 per cent., or 4,488 square miles, being 2,875,238 acres, are fit for agricultural purposes, mostly requiring irrigation. The timber is set down at 10 per cent., or 1,149,094 acres. For grazing, the estimate is 30 per cent., and as barren, 35 per cent. The total available land in the area surveyed is thus set down at 65 per cent., being 11,670.6 square miles, or 7,469,114 acres.

Temperatures and Rainfall.

	CAMP APACHE.	FORT BOWIE.		CAMP GRANT.		CAMP LOWELL.		CAMP MCDOWELL.		CAMP MOJAVE.		CAMP VERDE.		FORT WHIPPLE. (PRESCOTT).		FORT YUMA. (YUMA CITY).	
	Tem- per- ature	Tem- per- ature	Rain- fall	Tem- per- ature	Rain- fall	Tem- per- ature	Rain- fall	Tem- per- ature	Rain- fall	Tem- per- ature	Rain- fall	Tem- per- ature	Rain- fall	Tem- per- ature	Rain- fall	Tem- per- ature	Rain- fall
	Deg.	Deg.	Inch.	Deg.	Inch.	Deg.	Inch.	Deg.	Inch.	Deg.	Inch.	Deg.	Inch.	Deg.	Inch.	Deg.	Inch.
July	60-104	71- 103	0.50	58- 109	1.70	39- 113	0.08	72-113	0.00	47- 118	0.00	48- 113	0.14	65- 91	1.56	69- 112	0.00
August	66- 88	64- 97	1.34	55- 102	5.20	46- 104	2.73	65-108	0.56	52- 116	3.80	58- 102	2.52	64- 85	4.78	71- 106	1.60
September	52- 92	67- 99	0.01	53- 99	2.50	52- 103	0.62	54-110	0.00	45- 108	0.00	41- 97	0.26	50- 82	0.30	59- 104	0.00
October	28- 92	42- 96	0.03	35- 100	0.46	21- 101	0.00	33-108	0.00	27- 105	0.00	21- 95	0.00	33- 81	0.00	48- 100	0.00
November	25- 81	33- 85	1.12	31- 81	3.38	30- 91	1.32	33- 99	0.21	36- 89	0.50	20- 74	0.74	29- 72	0.80	46- 86	0.00
December	6- 62	20- 70	2.02	21 82	1.75	25- 78	0.97	27- 83	4.70	29- 67	2.80	6- 57	3.26	10- 65	2.55	39- 61	0.64
January	6- 68	21- 67	2.33	20- 85	1.58	19- 78	1.76	24- 83	3.10	27- 70	0.19	5- 59	2.65	17- 67	5.51	37- 72	0.55
February	10- 65	20- 67	5.40	16- 80	2.87	21- 75	1.66	18- 78	2.86	29- 69	5.00	12- 60	2.05	10- 55	5.68	35- 70	0.85
March	18- 72	32- 79	1.50	28- 86	2.45	30- 79	1.19	31- 79	1.06	39- 80	0.20	19- 72	1.05	20- 65	3.56	40- 82	0.20
April	31- 88	32- 82	0.35	30- 93	0.58	34- 97	0.43	43- 97	1.30	54- 96	0.10	27- 87	1.48	34- 75	1.70	45- 95	0.00
May	38- 94	48- 100	0.00	30- 101	0.07	42- 103	0.07	43-105	0.30	63- 107	0.90	34- 102	0.08	41- 82	0.65	50- 102	0.00
June	57-101	67- 100	0.00	54- 105	0.00	44- 108	0.00	54-114	0.00	75- 111	0.00	43- 107	0.00	55- 88	0.00	66- 108	0.00
		—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		14.60		22.54		10.83		14.09		13.40		14.19		27.09		3.84	

At Florence, July, 1877, the thermometer stood at 100 to 115 deg.

In Gila Valley, near the mouth of San Pedro, August and September, 1876, 50 deg. at sunrise; 105 deg. at 2 P. M.; early in October, 30 deg. at sunrise—90 deg. at 2 P. M.; close of October, 15 deg. at sunrise—90 deg. at 2 P. M.

In Gila Valley, on the New Mexico and Arizona line, October 17th, 1876, at sunrise, 28 deg.—at base of mountain range, ten miles distant, 40 deg.; Oct. 18th, at sunrise, 14 deg.—4,500 feet altitude; Oct. 19th, at sunrise, 40 deg.—5,200 feet altitude.

At Mineral Peak, during June and July, 1877, the thermometer repeatedly reached 100 deg.

August 8th to 13th, 1877, 30 miles below Sunset Crossing, on the Colorado Chiquito, the thermometer stood, at sunrise, 62 to 64 deg.; at 2 P. M. 82 to 85 deg.; at sunset, 71 to 75 deg.—3,700 feet above sea level.

A copy of this table as an image can be found [here](#).

Table of Altitudes—Principal Points in Arizona.

PLACE.	Lat.			Long.			Altitudes above Sea Level	REMARKS.
	°	'	"	°	'	"		
Antelope Springs							8,065.1	
APACHE CAMP	33	48	18.70	32	52		5,000.9	
Apache Mesa							5,800.0	
Art-too-hah (Cañon Creek)	35	44	43.28					Bunch-grass, cedars.
Beaver Creek	34	44	02.52				3,671.4	
Big Hills	33	23	07.70				5,702.5	
Bill William Mt.							8,000.0	
Black Hills or Tonto Plateau							9,000.0	Estimated.
Bonché's Fork	34	33	08.54				5,820.1	
Bowie Camp	32	10	16.02				4,871.6	
Bradshaw City							7,000.0	
Cedar Creek	34	04						
Chevelon's Fork							4,000.0	Trib Col. Chiquito.
Chevelon's Fork, upper course							5,200.0	Estimated.
Chiricahua Mts.							7,000.0	Estimated.
Desert Station	32	30	08.80				2,135.2	
Diamond Creek	35	45	19.11				1,350.4	
Disaster Rapids	35	55	52.10					Colorado River.
Eureka Springs							4,900.0	
FLORENCE	33	02	32.53					Town 1,000 inhabitants in Gila Valley, fertile.
Gila River near Camp Goodwin							2,517.0	
Graham Peak, Mt. Graham							10,516.0	Nearly 6,000 feet above base.
Grant, Camp (new)	32	25		32	23	10	3,985.0	Suily.
							4,753.0	Rothrowp.
							4,833.0	Wheeler Map.
							5,400.0	
Grant, Camp (old)	32	47	35.00	113	37	15	2,500.0	Abandoned, Jan. 1873.

Green Springs	36 11 13.00	4,931.2	Little wood & water, wood in gulch.
Limestone Water Pocket	36 32 18.40	5,405.4	Little wood, grass; to right of road, water in small gulch.
LOWELL CAMP (Tucson)	32 12 33 49	2,530.0	
McDOWELL CAMP	33 40 111 40	1,800.0	
MINERAL PARK		3,000.0	Approximate.
Moen-copie Cañon	36 08	4,984.1	
Mogollon Mesa		7,000.0	
MOJAVE CAMP	35 24 114 34 40	600.0	
Navajo Spring	36 46 19.10	4,410.02	Wood, grass, & water scarce.
New Creek (Ives)	35 36 51.00		Pabroach Spring; plenty of wood, water, and grass.
Nelson's Tanks	34 46 20.42	6,216.0	Or Mogollon Mesa.
Oraybe	35 52 57.00	4,756.8	Moqui Village and tank near.
Pah-guhn Springs	36 24 51.83		
Pah-wash	35 36 51.00		New Creek of Ives.
Paria River Cañon (Great bend Col.)	36 59	3,873.5	
Peach Orchard	35 46 42.40	6,297.5	
Picacho Station	32 44 20.67	1,750.2	
Picket Post	33 17 01.27	2,669.6	
Pinal Camp	33 21 01.45		
Pinal Creek	33 32	3,112.2	
Pinal Mountains	33 23 10.24	3,925.5	
Portage Rapids	35 48 35.90		
PRESCOTT	34 29.06 35 27 30	5,318.0	
Prieto Crossing	33 33 47.30	5,332.8	
Pueblo Colorado	35 42 10.40	6,400.9	
PUEBLO VIGA	32 49 00.00	2,711.6	
Puerto River (mouth)	34 53 16.80	5,083.0	
Rattlesnake Cañon	34 55 49.83	4,600.0	
Relief Springs	35 08 34.28	5,526.6	Near Lookwood's.
San Francisco Mts. (Humphrey's Peak)		12,561.0	Forest and sheep range. Extinct crater.
San Francisco Mts. average height		7,000.0	Volcanic.
San Pedro River	32 43	5,874.5	Grass, water, wood; settlements.
Santa Rita Mts.		8,000.0	Water, grass, wood plenty. Camp Toltec near Tubac.
Sierra Blanca		11,388.0	On Mexican line.

Sunset Camp	33	13	24.00		5,276.2	Colorado Chiquito.
Sunset Crossing	33	59	41.70		4,891.5	Colorado to Chiquito.
Truxton Springs	35	24	52.51		3,885.5	Bunch-grass, sage brush; water good.
TUCSON					2,500.0	Town, 4,000 inhabitants.
Verde, Camp	34	33		34 57	3,500.0	
Whipple, Fort	34	29	6.00	35 27 30		Near Prescott.
Willow Spring					7,195.0	
Wingate, Fort, N. M.	35	20		31 22	6,822.0	United States Post.
Wrightson, Mt. (Santa Rita)					10,500.0	Mining camps near.
Young's Spring	35	32	04.28			
Yuma, Fort	32	23	3.00	37 33 9	267.0	110 ft above river bank on bluff. Yuma city on east back of Colorado.
Zuni Mts. (N. M.)					9,000.0	Estimated.

The surveyors for the Atchison, Topeka & Santa Fé Railroad have pressed work beyond Cimarron, New Mexico, and expect to extend their surveys as far as Tucson, Arizona, during the present winter. The building of the road, however, will depend largely upon the corporation's ability in obtaining a land (or other) subsidy from Congress.

**List of all Mining Companies who have Filed their Articles of
Incorporation in the Office of the Secretary of Territory to October 1st,
1877.**

NAME	Capital Stock.	No. of Shares.	District.	County.	Principal Place of Business.
Arizona Chief	\$10,000,000	100,000	San Francisco	Mohave.	San Francisco.
Arizona Con	5,000,000	100,000	Peck	Yavapai.	Prescott.
Athens	10,000,000	100,000	Pioneer	Pinal.	San Francisco.
Bonanza King	10,000,000	100,000	Harcuoar?	Yuma.	"
Bronknow Con. M. & M.	1,800,000	18,000		Pima.	"
Cedar Valley	1,000,000	10,000	Cedar Valley	Mohave.	Mohave.
Cedar Valley G'd	2,500,000	25,000	Greenwood	"	San Francisco.
Cerbat Con. G. & S.	4,000,000	40,000	Wallapai	"	"
Champion Con. G. & S.	4,000,000	40,000	"	"	"
Colorado River Copper & Gold.	5,000,000	50,000		Yuma	"
Coronado	60,000	600	Arizona	and	New Mexico.
Cosmopolitan	10,000	100	At large		Prescott.
Cupel & Tiger	4,900,000	49,000	Wallapai	Mohave.	San Francisco.
Daisy Deane	3,000,000	30,000	At large		"
Detroit Copper	500,000	20,000	"		Detroit, Mich.
Eureka Bonanza	10,000,000	100,000	Eureka		San Francisco.
Excelsior Silver	10,000,000	100,000		Pinal.	"
Goodwin	10,000,000	100,000	Turkey Creek	Yavapai.	"
Greenwood Gold	2,500,000	25,000	Greenwood	Mohave.	"
Haskin	10,000,000	100,000	Globe	Pinal.	"
Keystone, 1 & 2, G. & S.	3,000,000	30,000	Wallapai	Mohave.	"
Lone Star G. & S.	50,000	5,000	"	"	Mohave Co.
Longfellow Copper	50,000	100	Arizona	and	New Mexico.
May Bean	2,500,000	100,000	Peck	Yavapai.	Prescott.
McCrackin Con	20,000,000	200,000		Mohave.	San Francisco.
McMillen	10,000,000	100,000	Globe	Pinal.	Santa Rosa, Cal.
Mina Madre	300,000	3,000	At large		Tucson.
Mineral Park (mill)	1,200,000	12,000	"		San Francisco.
Montour	96,000	96,000	"		"

Northern M. & M.	10,000,000	100,000	Pioneer	Pinal.	”
Ostrich M. & M.	300,000	6,000		Pima.	Tucson.
Peck	10,000,000	100,000	Peck	Yavapai.	San Francisco.
Pima G. & S.	10,000,000	100,000		Pima.	”
Pine Flat	100,000	5,000	Turkey Creek	Yavapai.	Prescott.
Silver King North	10,000,000	100,000	Pioneer	Pinal.	San Francisco.
Silver King South	10,000,000	100,000	Pioneer	”	”
Sixty-Three G. & S.	4,000,000	40,000	Wallapai	Mohave.	”
Tiger Silver	2,400,000	24,000	Tiger	Yavapai.	”
Tip Top	480,000	48,000		”	”
Vulture	5,000,000	50,000	Wickenburg	Marico’a.	New York City.
Wallace	5,000,000	50,000	Peck	Yavapai.	Prescott.
Wheatfield	2,400,000	240	At large		San Francisco.
Zalida	2,500,000	100,000	Lynx Creek	Yavapai.	Prescott.

Mining Districts in Yavapai County.

- | | | |
|-----------------|----------------------|-------------------|
| 1. Aqua Fria. | 8. Lynx Creek. | 14. Tiger. |
| 2. Black Cañon. | 9. Mineral Point. | 15. Verde. |
| 3. Big Bug. | 10. Mountain Spring. | 16. Wickenburg. |
| 4. Bradshaw. | 11. Pine Grove. | 17. Walnut Grove. |
| 5. Goodwin. | 12. Peck. | 18. Walker. |
| 6. Hassayampa. | 13. Turkey Creek. | 19. Weaver. |
| 7. Hum Bug. | | |

N. B.—A number of companies have also been incorporated in California and elsewhere, for the purpose of mining in Arizona.

Composition of Soils in Arizona.

(Accompanying Geological Report, Wheeler Expedition, 1875.)

	Moquis Villages.	Chevelon's Fork.	Mogollon Mesa.	San Francisco Mountains.	Rio San Pedro.	Camp Grant.	Rio Gila.
Predominating rock	Sandstone.	Sandstone.	Sandstone.	Basalt.	Granite and rhyolite.	Granite.	Basalt and rhyolite.
Sand	72.04	53.10	42.20	15.95	14.00	61.20	
Silt, with some clay	27.96	43.55	37.98	62.97	75.40	34.07	92.26
Potassa	0.072	0.092	0.115	0.130	0.401	0.131	0.242
Soda,	Traces.	0.010	Trace.	0.017	0.051	0.014	0.039
Lime	1.665	0.319	0.153	0.684	4.356	1.998	1.798
Magnesia			0.029	Trace.	1.019	0.203	0.570
Alumina	} 2.327	2.559	} 2.013	9.729	6.850	2.304	2.311
Oxide of iron							
Phosphoric acid	0.031	0.070	0.058	0.284	0.213	0.095	0.214
Sulphuric acid	Trace.	Trace.	Trace.	Trace.	Trace.	0.010	Traces.
Hydrosopic water	2.221	1.89	10.97	12.83	6.09	2.80	4.98
Chemically-bound water and organic matter	1,529	1.46	8.84	8.25	4.51	1.93	2.76
Insoluble in hydrochloric acid	94.60	93.55	77.81	71.09	71.10	87.52	84.85

Among the secondary precious stones that have been found in Arizona are black and green tourmalines, peridots, beautiful garnets of every tint, bloodstone, jaspers and agate of every character, while fire and white opals are found in certain localities. Zircona, in crystals, very minute, of the cube form, and in masses, has been discovered. Sufficient vein matter has not been found to determine any rich deposit of fine crystals.

Tabular Statement of Indian Affairs in Arizona,

Showing Population, Products, Stock, Education, etc., on the several Reservations, including the Navajo.

Names of Agency and Reservation.	Tribes.	Population.			Number following civilized pursuits	Percentage of means of subsistence from		Area of Reservation.	Acres Cultivated.	Produce raised during fiscal year ending June 30, '77		Stock Owned.		Number of Children at school.		Expenditure
		Males	Females	Total.		civilized pursuits	Hunting, Fishing & Gathering.			Bush. Wheat.	Bush. Corn.	Horses and Mules.	Cattle.	Males	Females	
Colorado River	Mojaves and Chemehuevis	610	530	820	424 Mojaves			128,000	470			110				
	Hualapais ¹			600												
	Coahuilas ¹			150												
	Cocopahs ¹			180												
Moquis Pueblos	Moquis	850	850	1,700	1,700	90	10	No reservation.	3,000	200	50			40		\$5,0
Pimas and Maricopas	Pimas ²	2,200	2,300	4,100	800	75	25	64,000	7,300	40,000	150	1,812	800	44	22	1,7
	Maricopas ²			400												
Papagoes	Papagoes ²	2,900	3,000	5,900	950	75	25	70,400	700	2,100	500	4,500	2,500	44	50	1,8
White Mountain Reservation, San Carlos	Pinal & Arivipais Apaches			1,051												
	Chiricahua Apaches			297												
	Mojave "			618	715	6	6	2,528,000	545	2,333	2,300			2	5	2
	Yuma "			352												
	Tonto "			629												
	Coyetoro "			1,612												
	Southern " ³	600	1,000	1,600												
Not under an agent or at reservation	Yumas			930												
	Mohaves			700												
Navajo (Arizona and New Mexico)	Navajoes ⁴	5,852	6,016	11,868	3,500	90		3,328,000	6,000	51,400	15,200	1,000	17	9		5
				33,847	8,089			6,118,400	18,015	54,460	62,212	4,300	147	86		\$9,2

¹ Not on reservation, but in charge of agent.

² The Papago Agency has recently been consolidated with that of the Pimas and Maricopas.

³ The Southern Apaches, consisting of the Gila, Mogollon, Mumbre, and Chiricahua Apaches, were removed from the Hot Springs Agency, in New Mexico, to the San Carlos Agency, in May, 1877, by Agent Clum.

⁴ The Navajo Agency is in both Arizona and New Mexico. The Navajoes mostly live in New Mexico.

A copy of this table as an image can be found [here](#).

The total areas of above reservations are 9,560 square miles; tillable acres, 46,000. Besides corn and wheat, 5,200 bushels of barley and oats, and 3,456 bushels of vegetables were raised; 380 tons hay and 800 cords wood were cut. The Navajoes sold woolen materials, principally blankets, to the amount of \$20,000. The number of church members recorded is 13.

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- Cœdex Borgian. Fac-simile, Borgian Museum, Rome. Vol. iii.
- Cœdex Boturini. Fac-simile, Collection of Boturini. Vol. i.
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Besides these authorities, files of the *Arizonian*, published at Tubac; of the *Citizen*, Tucson; the *Sentinel*, Yuma; the *Miner*, Prescott; and the *Enterprise*, of Mineral Park and Prescott; with the *Alta California* and *Evening Post*, San Francisco; have been copiously referred to. There were also reports of the signal officer in charge of U. S. Military Telegraph, of the department records, with private notes and journals of J. D. Graham, W. G. Boyle, Prof. Rickard and others, consulted during the preparation of this work.

Arizona Business Directory.

[The list is arranged according to post offices, and classified according to counties. It is not presented as an accurate and complete one, but it is the best that could be obtained, owing to the carelessness and indifference of those who should be most interested in forwarding the information solicited.]

Official Directory.

TERRITORIAL OFFICERS.

Delegate to Congress	Hiram S. Stevens
Governor	John P. Hoyt
Secretary	J. J. Gosper
Auditor	E. P. Clark
Treasurer	T. J. Butler
Surveyor-General	John Wasson
Supreme Judge, 1st District	Chief Justice French
” ” 2d ”	De Forest Porter
” ” 3d ”	C. A. Tweed
Clerk Supreme Court	Joseph B. Austin
United States District Attorney	E. B. Pomeroy
United States Marshal	W. W. Standifer
Deputy Marshal, 1st District	Wm. J. Osborn
Collector Internal Revenue	Thomas Cordis
Register U. S. Land Office, Prescott	W. N. Kelly
” ” ” Florence	Levi Ruggles
Receiver ” ” ”	C. D. Poston
” ” ” Prescott	George Lount

COMMISSIONERS OF DEEDS.

San Francisco:

J. H. Blood	July 17th, 1874
Ed. Chatten	January 7th, 1875
John H. B. Wilkins	February 3d, 1875
N. Proctor Smith	March 17th, 1876
A. S. Gould	April 22d, 1875
James Brooks	June 8th, 1875
E. V. Joice	June 5th, 1875
Samuel Herman	February 28th, 1876
Wm. Harney	February 12th, 1876
James H. Lawrence	June 17th, 1876
Samuel S. Murfey	June 23d, 1876
H. M. Morgan	August 1st, 1876
Frank V. Scudder	August 9th, 1876
J. E. Russel	October 13th, 1876
F. C. Wegener	December 13th, 1876
Lewis Franconi	January 26th, 1877
Holland Smith	April 9th, 1877
E. H. Long	May 1st, 1877
Henry C. Blake	May 7th, 1877
E. H. Tharp	May 3d, 1877
Charles J. Torbut	June 6th, 1877

Los Angeles:

H. Fleishman	February 17th, 1876
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San Diego:

Michael Keating	September 5th, 1876
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Sacramento:

Ed. Cadwalader	March 16th, 1876
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Alameda:

Will. H. Burrall	January 11th, 1877
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District of Columbia:

John C. Starkweather	July 13th, 1877
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Pedro Aguiere	March 7th, 1877
James Brooks	June 8th, 1875
E. V. Joice	June 5th, 1875
B. H. Hereford	March 9th, 1877
W. J. Ross	July 26th, 1877
H. B. Jones	November 1st, 1873
J. E. McCaffrey	January 26th, 1874
Hugh Farley	November 19th, 1874
Ed. A. Yerkes	March 19th, 1876
George Cooler	August 17th, 1876
T. H. Goodwin	August 17th, 1876
Hylar Ott	September 30th, 1876
H. S. Delos	November 3d, 1876
James Holden	January 18th, 1877
<i>Yavapai County, 6:</i>	
Ed. H. Wells	July 17th, 1875
Murat Masterson	March 1st, 1876
A. J. Bruner	August 6th, 1877
J. E. Roberts	June 11th, 1874
James Gough	February 5th, 1876
Henry A. Begalon	June 30th, 1877
<i>Yuma County, 4:</i>	
Peter Dohl	January 17th, 1876
H. N. Alexander	September 20th, 1876
P. M. Spinuello	May 18th, 1877
James Reilly	August 31st, 1877
<i>Maricopa County, 3:</i>	
William A. Hancock	September 2d, 1875
H. E. Lacy	November 5th, 1875
J. A. Parker	February 5th, 1876

Mojave County—None
qualified.

Pinal County—None qualified.

Bank of Arizona.

DIRECTORS.

Joseph Brandenstein, of A. S. Rosenbaum & Co.	San Francisco
Joseph May, of Livingston & Co.	San Francisco
Samuel Lewis, of Lewis Bros.	San Francisco
Sol. Lewis	Prescott
M. W. Kales	Prescott
John J. Gosper	Prescott
T. J. Butler	Prescott

CORRESPONDENTS.

The Anglo-California Bank (Limited)	San Francisco
Messrs. J. & W. Seligman & Co.	New York

Agua Fria, Yavapai Co.

N. B. Bowers, rancher.
Fred. Hildebrand, rancher.
W. J. Herman, rancher.
John Mario, rancher.
D. Miller, cattle dealer.
Fred Plum, rancher.
John Reese, rancher.
H. Spaulding, rancher.

Fred. Williams, rancher.

Alexandra.

Anders & Rowe, general merchants.

Henry A. Bigelow, Notary Pub. and conveyancer.

James C. Reynolds, prop. Summit House.

Summit House (J. C. Reynolds, prop.)

Allen.

Berardo Freyde, merchant and hotel keeper.

J. P. Heyward, general merchant.

S. G. Ladd, Justice of Peace.

John McLaws, Postmaster.

Joseph H. Richards, wagon-maker and builder.

Camp Grant, Pinal Co.

Norton & Stewart, post-traders.

Stevens & Teague, station-keepers.

M. W. Stewart, Postmaster.

T. A. Toney, attorney at law.

M. L. Wood, proprietor Hotel de Luna.

Camp Thomas, Pima Co.

Harry Campbell, hair-dresser, &c.

Clanton House (J. H. Slinkard, prop.)

Hellar & Mullett, wagon-makers and blacksmiths.

Leitch & Kehoe, cattle-dealers and butchers.

Lamb & Webber, merchant tailors.

Overton & Luten, livery and sale stable.

J. W. Powell & Co, general merchants.

F. Staples & Co, wholesale merchants.

Wm. Eagle, druggist.

Valley House (Mrs. Alice Meyer.)

Camp Verde, Yavapai Co.

C. P. Head & Co, general merchants.

Cienega, Pima Co.

W. W. Compton & Co, mining, etc.

George W. Hand, Postmaster and station agent.

Cerbat, Mojave Co.

Caldwell Wright, County Recorder and searcher of records.

Clifton, Yavapai Co.

Eusebio Casteñeda, attorney at law.

Coronado Copper Mining Co, Wm. Grant, supt.

William Grant, Postmaster.

Joy Copper M. Co, Wm. McCormick, supt.

Longfellow Copper M. Co, Wm. Smith, supt.

Longfellow House (Pablo Salcido, prop.)

Wm. Smith, mining and civil engineer.

Louis Smadbeck, M. D., physician.

Thompson Copper M. Co, E. V. Thompson, supt.

Desert Station.

(48 miles from Florence.)

Chas. H. Labaree, prop.

Ehrenberg, Yuma Co.

J. Goldwater, merchant.
R. E. Farington, freighter.

East Phoenix, Maricopa Co.

C. H. Veil, Salt River Flouring Mill.
John J. Devine, Clerk of Court, and Recorder.
Elliott House (J. C. Soss & Bro, prop.)
Morris F. Emerson, notion store.
Florence Hotel (P. S. Emprey, prop.)
M. Fearney, supt. Stonewall Jackson M. Co.
H. Goldberry & Co, general merchants.
J. Guindain & Co, general merchants.
Francisco Padrias, tin shop.
Col. Chas. D. Poston, Register U. S. Land Office.
Clement Romo, general merchant.
Col. Levi Ruggles, Receiver U. S. Land Office.
Charles Summers, M. D., physician.
H. B. Summers, attorney at law.
Charles Whitlow, supt. Silver Bell M. Co.
J. D. Walker, Probate Judge.
Peter Will & Co, brewery.

Florence, Pinal Co.

Blunt, M. D., physician.
S. B. Chapin, M. D., physician and druggist.
Peter R. Boady, flouring mill.
Bushart & Co, flouring mill.
J. P. Clum, attorney at law.
Citizen, weekly paper, J. P. Clum, Editor.
M. Calisher, general merchant.
Joseph Collingwood, Postmaster.
J. Collingwood & Co, wholesale merchants.
Geo. A. Brown, agent Wells, Fargo & Co.
P. S. Empey, hotel.

E. Van Hasslach, M. D., physician.
John W. Leonard, attorney at law.
H. B. Summers, attorney at law.
W. S. Edwards, civil engineer and surveyor.
Wm. J. Osborn, atty. at law and notary public.
Thomas McLellan, house and sign painter.
Miners' Exchange, (Jack Upton, prop.)
J. DeNoon Reymert, attorney and counsellor at law.
W. H. Merritt, assayer
Elliott House, (J. C. Loss & Bro., proprietors.)
M. Calisher, general merchandise.
Florence Wine Rooms, (Lew Baitey, prop.)
Florence Meat Market, (L. Reney, prop.)
Calhoun & Bush, livery stable.
C. F. Palmer, machinist.
City Bakery, Hy. Watzlarzick.
Samuel Bostick, barber and hair-dressing.
Florence Hotel, (P. S. Empey, prop.)
Jos. Messinger, gen'l merchandise.
Florence Bakery, G. Bonerand.
Davis & Young, stoves and tinware.
J. Guindani & Co., general merchandise.

Fort Whipple, Yavapai Co.

Peter B. Brannan, post trader and general merchant.

Gila City, Yuma Co.

A. Lange, prop, station.

Globe City, Pinal Co.

Shuyrock & Hise, agents Silver City and Globe Express Line.

Greenwood, Mojave Co.

Greenwood Express, Ehrenberg to Greenwood, John Doling, prop.

Hayden's Ferry, Maricopa Co.

Chas. F. Hayden & Co, general merchants.

Hackberry, Mojave Co.

Hackberry Mill and M. Co.

L. B. Seaver, supt. Hackberry Mill and M. Co.

McCrackin Hill, Mojave Co.

Felix Simon, ranch and station.

McDowell, Maricopa Co.

John Smith, general merchant.

Mineral Park, Mojave Co.

James P. Bull, Commissioner Deeds California and Nevada.

Louis Cazino, restaurant.

Alonzo E. Davis, attorney at law.

James B. Lawrence, physician.

Mohave City, Mohave Co.

Paul Breon, Postmaster.

Breon & Spear, general merchants.

Edwin Dane, blacksmith.

S. A. Freeman, physician.

Lambert & Schimps, butchers.

Mods Mine, Peter Ortiz, supt.

A. A. Spear, freighter and contractor.

Montezuma Store.

(12 miles below Florence.)

Austin & Dempsey, gen'l merchandise.

Phoenix, Maricopa Co.

G. H. Ansy, attorney at law.
John L. Alsop, attorney at law.
John Burger, wagon-maker.
J. M. Catten, saloon.
Julius Baneslen, baker.
Ellis Brothers, general merchants.
Wash Evans, blacksmith.
John George, saloon.
A. Goldman, general merchant.
J. J. Gardner, hotel.
W. A. Hancock, attorney at law.
Chris Heiman, bakery.
Thomas How, furniture mftr.
E. Irvine & Co, general merchants.
John Lentgerding, blacksmith.
—— Liemon, attorney at law.
Leasing & Leaskin, general merchants.
J. D. Monihan, livery stable.
H. Morgan & Co., general merchants.
Murphy & Dennis, freighters, etc.
J. E. Naylor, attorney at law.
M. L. Peralta, general merchant.
R. R. Pearson, carpenter and builder.
Thomson & Thibido, druggists.
J. W. Thomson, carpenter and builder.
Henry Sayers, saloon.
M. Wormser, general merchant.
John West, blacksmith.
Charles White, hotel keeper.

K. S. Woolsey, flouring mill.
C. H. Veil, flouring mill.

Picket Post, Pinal Co.

H. DeGroat, Jr., assayer.

Prescott, Yavapai Co.

Anders & Row, general merchants.
ARAM, M'CUSKER & BRIMER, attorneys at law.
ARIZONA BREWERY, Jul. N. Rodenburg.
BANK OF ARIZONA.
L. BASHFORD & CO, wholesale general merchants.
George T. Becker, baths and hair-dressing.
G. A. Behlow, tobacco and cigar dealer.
E. J. BENNETT, C. E., County Surveyor,
City Engineer, and Deputy U. S. Mineral Surveyor.
H. A. Bigelow, notary public and conveyancer.
Blackburn & Schulze, city brewery.
Blake & Co., assayers.
Bones & Spencer (Chas. L. Spencer and James E. Bones), dealers in
tobacco, cigars, stationery, etc.
THOS. C. BRAY, general merchant.
Fred. G. Brech, wagon and blacksmith shop.
GIDEON BROOKE, prop. Plaza Feed and Sale Stable.
WILLIAM M. BUFFUM, general merchant.
Cabinet Chop House (D. C. Thorne, prop.)
California Meat Market (W. Coffee, prop.)
CALIFORNIA AND ARIZONA STAGE COMPANY, J. H. Hughes,
President.
CAMPBELL & PARKER, architects and builders.
H. H. Cartter, Probate Judge, Justice of the Peace and conveyancer.
C. F. Cate, prop. Nifty saloon.
Clipper Mill (Jas. G. Wiley, prop.)
Cohen, Casey & Co., prop's., O. K. feed and sale stables and yard.

Robert Connell, prop. Champion saloon.
Leon Cordier, baths and hair-dresser.
Charles Crocker, boots and shoes.
George W. Curtis, saw-mill.
Dr. Warren E. Day, physician and surgeon.
DUDLEY HOUSE (Fred Williams, prop.)
T. J. DRUM, attorney at law.
Emanuel & Vallony, prop'rs restaurant and saloon.
J. L. FISHER, real estate agent, auctioneer, etc.
FITCH & CHURCHILL, attorneys and counsellors at law.
J. GOLDWATER & BRO., wholesale and retail merchants.
G. E. GOODFELLOW, physician and surgeon.
F. L. B. Goodwin, attorney at law.
Hall & Young, house and sign painters.
J. P. Hargrave, attorney at law.
Guilford Hathaway, Gray Eagle stables.
Charles T. Hayden & Co., (Charles T. Hayden and Hezekiah Brooks)
 general merchandise.
C. P. HEAD & CO., wholesale merchants.
R. H. Hereford & Briggs Goodrich, attorneys.
JOHN HILL, stone cutter, contractor and builder.
JOHN HOWARD, attorney at law.
W. J. HOWARD, watchmaker and jeweler.
JAMES HOWEY, general blacksmith.
Kearney's Saloon.
Jay G. Kelley, assayer and metallurgical chemist.
Kelly & Stephens, (Wm. N. Kelly and F. A. Stephens) newsdealers and
 general merchants.
George D. Kindall, prop. Pioneer drug store.
Philip Kohlheyer, prop. City meat market.
J. A. Lewis, architect, builder, contractor, etc.
H. Loisilbon, merchant.
LOWELL & CROUCH, practical roofers.
J. N. McCANDLEN, physician and surgeon.
Murat Masterson, attorney at law.
S. C. Miller & Bro., forwarding merchants.
Milligan & Co. meat market.

D. F. MITCHELL, prop. Capital Art Gallery (photographic.)
Montgomery House (Whitehead & Plumridge, props.)
M. H. Mooney, merchant tailor.
Benjamin Morgan, attorney and counselor-at-law.
Morgan & Buck, baths, and hair-dressers.
MORGAN & DOUGHERTY, jewelry, watches and engraving.
Mount Hope Saw-mill, (Byron Sherman & Co., props.)
Mount Hope, near Prescott.
A. O. Noyes, Pioneer saw-mill.
"O K" STABLE, camp house and feed yard.
T. W. Otis, general merchant and agent Singer Sewing Machine.
Pacific Brewery, (John Raibb, proprietor.)
PLAZA FEED AND SALE STABLE (Gideon Brooke, proprietor.)
Pioneer Drug Store (George D. Kendall.)
Pioneer Furniture Store (Ed. Stahl, proprietor.)
Pioneer House (Ganz & Hatz, managers.)
Pioneer Saw-Mill (Caldwell & Pardee, proprietors.)
Pioneer Store, (John G. Campbell) general merchandise.
GEORGE B. PORTER & CO., furniture, etc.
John Raible, Pacific brewery.
Paul J. Robert, attorney and counselor-at-law.
J. M. Rodenberg, Arizona brewery.
C. F. Rogers & Co. (Prescott meat market.)
Rush & Wells, (John A. Rush and Ed. W. Wells) attorneys and counselors-at-law.
Sazarac Saloon.
Frank Schunemann, blacksmith and wagon-maker.
Smith & Hawley, carpenters and builders.
C. J. Spencer & J. E. Bones, tobacco and stationery.
A. J. Thibods, M.A., M.D., physician and surgeon.
E. Thiele, M.D., physician, surgeon and accoucher.
D. C. Thom, Cabinet chop-house.
J. H. Thrans & W. H. Reardon, architects and builders.
B. J. Wade, attorney at law.
GEORGE M. WATERS, millwright and contractor.
B. H. Weaver, general merchant.
Paul Weber, attorney and counselor-at-law.

W. W. Weed, official short-hand reporter.
William Wilkerson, (Clerk to Board of Supervisors Yavapai County.)
Wilson & Haskell, (W. Z. Wilson and A. S. Haskell) sash, door and blind
factory.
J. G. Worthington, sign and carriage painter and trimmer.
G. Urfer, lodging-house.

Skull Valley.

A. J. Whitehair, general merchant.
J. H. Dickson, station house.

Silver King, Pinal Co.

F. M. Sponagle, physician and surgeon.

Sahuarita Ranch, Pima Co.

Roddock & Brown, hotel station.

Tres Alamos.

Leonardo Apodaca, rancher.
A. Blair, rancher.
Jesus Dias, general merchant.
Thomas Dunbar, Postmaster and station-keeper.
G. Gibson, rancher.
C. M. Hooker, merchant and hotel keeper.
G. King, physician.
Frank Long, rancher.
Levi Pitts, U. S. mil. tel. operator.
A. A. Wilt, merchant.

Tubac.

J. Lillie Mercer, merchant.

Tucson.

Abadie & Co., saloon.

Juan N. Acuna, agt. Sonora & Arizona Stage Line.

John B. Allen, Mayor, merchant.

J. H. Archibald, general merchant.

J. M. Berger, jeweler.

GEORGE A. BOWEN, agt. Southern Pac. Mail Line.

Thomas Belknap, blacksmith.

C. O. Brown, saloon.

Henry Buchman, photographer.

S. W. CARPENTER, County Recorder and Notary Public.

James Carroll, livery and sale stable.

George Cooler, saloon.

Cosmopolitan Hotel, (Otto Reichenbach, prop.)

Davis & Young, hardware, stoves, etc.

Dill & Holt, "Cabinet" saloon.

C. R. Drake, Asst. Postmaster.

S. H. Drachman, general merchant.

C. T. Etchell, blacksmith.

FARLEY & POMEROY, attorneys at law.

Geo. Forster, saloon.

Thomas Gardner, lumber merchant.

J. Goldberg, American grocery.

John Goldtree, Telegraph saloon.

T. F. Griffith, livery stable.

Transcriber's Notes

Added periods between the dollar and cents in the tables on [page xvii](#) and [page xi](#).

The reprint that was used as the source for the ebook was flawed, and two missing words were added to the notes on [page i](#), from an alternate document: "to the *amount of* \$20,000."

Obvious typographical errors have been silently corrected. Variations in hyphenation and accents have been standardised but all other spelling and punctuation not mentioned here remains unchanged.

To aid readability on smaller devices, images of some of the more complex tables have been provided at the end of the book.

Images of Tables

Image of [Rainfall](#) table.

Temperatures and Rainfall.

CAMP APACHE.	FORT BOWIE.	CAMP GRANT.	CAMP LOWELL.	CAMP MCDOWELL.	CAMP MOJAVE.	CAMP VERDE.	FORT WHIPPLE. (PRESCOTT).	FORT YUMA. (YUMA CITY).									
Tem- per- ature	Tem- per- ature	Tem- per- ature	Tem- per- ature	Tem- per- ature	Tem- per- ature	Tem- per- ature	Tem- per- ature	Tem- per- ature									
Deg.	Deg.	Deg.	Deg.	Deg.	Deg.	Deg.	Deg.	Deg.									
Rain- fall	Rain- fall	Rain- fall	Rain- fall	Rain- fall	Rain- fall	Rain- fall	Rain- fall	Rain- fall									
Inch.	Inch.	Inch.	Inch.	Inch.	Inch.	Inch.	Inch.	Inch.									
July	60-104	71-103	0.50	58-109	1.70	39-113	0.08	72-113	0.00	47-118	0.00	48-113	0.14	65-91	1.56	69-112	0.00
August	66-88	64-97	1.34	55-102	5.20	46-104	2.73	65-108	0.56	52-116	3.80	58-102	2.52	64-85	4.78	71-106	1.60
September	52-92	67-99	0.01	53-99	2.50	52-103	0.62	54-110	0.00	45-108	0.00	41-97	0.26	50-82	0.30	59-104	0.00
October	28-92	42-96	0.03	35-100	0.46	21-101	0.00	33-108	0.00	27-105	0.00	21-95	0.00	33-81	0.00	48-100	0.00
November	25-81	33-85	1.12	31-81	3.38	30-91	1.32	33-99	0.21	36-89	0.50	20-74	0.74	29-72	0.80	46-86	0.00
December	6-62	20-70	2.02	21-82	1.75	25-78	0.97	27-83	4.70	29-67	2.80	6-57	3.26	10-65	2.55	39-61	0.64
January	6-68	21-67	2.33	20-85	1.58	19-78	1.76	24-83	3.10	27-70	0.19	5-59	2.65	17-67	5.51	37-72	0.55
February	10-65	20-67	5.40	16-80	2.87	21-75	1.66	18-78	2.86	29-69	5.00	12-60	2.05	10-55	5.68	35-70	0.85
March	18-72	32-79	1.50	28-86	2.45	30-79	1.19	31-79	1.06	39-80	0.20	19-72	1.05	20-65	3.56	40-82	0.20
April	31-88	32-82	0.35	30-93	0.58	34-97	0.43	43-97	1.30	54-96	0.10	27-87	1.48	34-75	1.70	45-95	0.00
May	38-94	48-100	0.00	30-101	0.07	42-103	0.07	43-105	0.30	63-107	0.90	34-102	0.08	41-82	0.65	50-102	0.00
June	57-101	67-100	0.00	54-105	0.00	44-108	0.00	54-114	0.00	75-111	0.00	43-107	0.00	55-88	0.00	66-108	0.00
		14.60		22.54		10.83		14.09		13.40		14.19		27.09		3.84	

Showing Population, Products, Stock, Education, etc., on the several Reservations, including the Navajo.

Image of [Statement of Indian Affairs](#) table.

*** END OF THE PROJECT GUTENBERG EBOOK ARIZONA GHOST
TRAILS ***

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